Spatial Data Analysis Beyond Interpolation:

A Functional Analytic Perspective on Geostatistics, Machine Learning and Geospatial Foundation Models

Abstract

This chapter presents a unified perspective on spatial data analysis that extends beyond classical interpolation. Spatial observations are treated as finite and often irregular samples from underlying continuous fields, implying that spatial dependence, support, scale, and uncertainty must be addressed explicitly rather than regarded as secondary considerations. The chapter contrasts two major modeling traditions: geostatistics, which represents spatial structure through covariance and semivariogram models, and machine learning, which emphasizes flexible nonlinear prediction from environmental covariates but often does not model spatial dependence directly. A functional analytic framework is used to show that both approaches can be interpreted as approximation problems defined on structured function spaces. Under this perspective, kriging, Gaussian processes, kernel methods, regularization, inverse problems, and operator learning emerge as closely related formulations within a common mathematical foundation. This paper further examines uncertainty quantification, change-of-support effects, multiscale spatial representation, multivariate data fusion, and hybrid approaches that combine geostatistical and machine learning concepts. A synthetic spatial case study comparing Random Forest, Ordinary Kriging, and regression kriging demonstrates how hybrid models can exploit both nonlinear covariate relationships and residual spatial dependence, yielding improved predictive performance over either framework alone. Recent geospatial foundation models introduce a new paradigm in which representations are learned from massive Earth Observation archives. I argue that these models can be interpreted as learned spatial operators acting on structured function spaces. Under this perspective, foundation models, geostatistics, Gaussian processes, kernel methods, and neural operators become closely related approximations within a common functional analytic framework. The central message is that robust spatial inference requires not only accurate prediction, but also explicit treatment of spatial dependence, support, scale, and uncertainty, motivating integrated methodologies that combine statistical, computational, and physical representations of spatial systems.

Keywords: spatial statistics, geostatistics, kriging, spatial autocorrelation, machine learning, uncertainty modeling, data fusion, support analysis

Contents

1 Introduction

Spatial phenomena are intrinsically continuous in space and time. Observed datasets therefore represent finite and irregular samples of an underlying spatial field rather than independent observations [23, 30].

Let

$$f : D \subset \mathbb{R}^d \to \mathbb{R} \tag{1}$$

denote a spatial process defined over a spatial domain D. The observed dataset

$$\{f(\mathbf{s}_1), \dots, f(\mathbf{s}_n)\} \tag{2}$$

is only a finite sampling of this continuous object.

From this perspective, spatial modeling becomes an approximation problem defined over functional spaces. The central question is not merely how to interpolate points, but how to infer a spatially structured function from incomplete and noisy observations.

Spatial datasets possess several properties that distinguish them from conventional tabular data. Nearby observations are often more similar than distant observations, directional effects may arise, sampling locations are irregular, and the support of the observations can vary substantially across sensors and measurement protocols [52, 122].

These characteristics imply that spatial dependence is not a secondary technical detail. It fundamentally alters inference, prediction, validation, and uncertainty quantification. Ignoring spatial structure can produce biased parameter estimates, misleading validation scores, and maps with little statistical interpretability.

Historically, two major frameworks have dominated spatial prediction.

The first is geostatistics, which models spatial dependence explicitly through covariance or semivariogram structures [30, 85]. The second consists of machine learning approaches, which emphasize flexible nonlinear prediction from explanatory variables but often do not explicitly represent spatial covariance structure [58].

These frameworks are often presented as competing paradigms. Mathematically, however, both can be interpreted as approximation procedures defined over functional spaces. Geostatistics emphasizes stochastic dependence, uncertainty propagation, and spatial covariance, whereas machine learning emphasizes nonlinear operator approximation in high-dimensional predictor spaces.

This chapter adopts a functional analytic perspective to establish a common mathematical foundation for both approaches. Spatial fields are interpreted as elements of infinite-dimensional spaces, kriging is interpreted as an orthogonal projection problem, and covariance functions are interpreted as positive-definite kernels.

Under this interpretation, geostatistics and machine learning become closely related through reproducing kernel Hilbert spaces, operator theory, regularization methods, and inverse problem formulations.

The remainder of the chapter is organized as follows. Section 2 introduces the functional analytic foundations of spatial modeling. Section 4 formalizes spatial dependence through covariance operators and stochastic fields. Subsequent sections compare geostatistical and machine learning approaches for spatial prediction, uncertainty quantification, multiscale analysis, and hybrid spatial modeling.

2 Functional analytic foundations of spatial modeling

2.1 Spatial fields as elements of function spaces

Spatial analysis is often introduced through maps, interpolation, and covariance functions. At a deeper mathematical level, however, most spatial models can be formulated as problems defined over functional spaces [77, 102].

Let

$$D \subset \mathbb{R}^d \tag{3}$$

denote a spatial domain and let

$$f : D \to \mathbb{R} \tag{4}$$

represent a spatial field.

Rather than interpreting the observations merely as discrete samples,

$$\{f(\mathbf{s}_1), f(\mathbf{s}_2), \dots, f(\mathbf{s}_n)\}, \tag{5}$$

functional analysis interprets the spatial process itself as an element of an infinite-dimensional function space [64, 94].

This viewpoint is natural in environmental sciences because variables such as temperature, precipitation, salinity, soil moisture, pollutant concentration, and biomass are intrinsically continuous. The observations are finite measurements of an underlying spatial continuum rather than isolated quantities.

Many spatial processes can therefore be represented in Banach spaces such as

$$L^p(D), \tag{6}$$

equipped with norm

$$\|f\|_p = \left(\int_D |f(\mathbf{s})|^p d\mathbf{s} \right)^{1/p}. \tag{7}$$

[13].

The choice of norm determines which properties of the field are emphasized. For example, L^2 spaces emphasize energy and variance, while Sobolev spaces additionally incorporate smoothness and differentiability constraints [1].

2.2 Banach and Hilbert spaces

A Banach space is a complete normed vector space. Completeness ensures that convergent sequences remain inside the space, which is essential for the existence and stability of solutions in approximation and inverse problems [28].

Among Banach spaces, Hilbert spaces play a central role in spatial statistics because they are equipped with an inner product:

$$\langle f, g \rangle = \int_D f(\mathbf{s}) g(\mathbf{s}) \, d\mathbf{s}. \tag{8}$$

This induced geometry permits the definition of orthogonality, projection operators, spectral decompositions, and reproducing kernels [77].

Most classical geostatistical methods can be interpreted naturally inside Hilbert spaces of square-integrable random variables [23, 30].

2.3 Projection theory and spatial estimation

Let $\mathcal{H}$ denote a Hilbert space of random variables. Spatial prediction can then be formulated as an orthogonal projection problem.

Suppose observations are available at locations

$$\{\mathbf{s}_1, \ldots, \mathbf{s}_n\}. \tag{9}$$

The predictor of the unobserved quantity

$$Z(\mathbf{s}_0) \tag{10}$$

is obtained by projection onto the subspace generated by the observations:

$$\mathcal{H}_n = \operatorname{span}\{Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_n)\}. \tag{11}$$

The projection estimator is therefore

$$\hat{Z}(\mathbf{s}_0) = P_{\mathcal{H}_n} Z(\mathbf{s}_0), \tag{12}$$

where $P_{\mathcal{H}_n}$ denotes the orthogonal projection operator.

Under second-order assumptions, ordinary kriging emerges naturally as the minimum-variance linear projection estimator [30, 84].

The kriging variance corresponds to the norm of the residual projection error:

$$\sigma_K^2 = \|Z(\mathbf{s}_0) - \hat{Z}(\mathbf{s}_0)\|_{\mathcal{H}}^2. \tag{13}$$

This interpretation is deeper than the traditional weighted-average formulation because it directly connects kriging with approximation theory, operator theory, and stochastic processes [104, 108].

2.4 Covariance kernels and reproducing kernel Hilbert spaces

Covariance functions are central objects in spatial modeling:

$$C(\mathbf{s}_i, \mathbf{s}_j) = \operatorname{Cov}\left[Z(\mathbf{s}_i), Z(\mathbf{s}_j)\right]. \tag{14}$$

From a functional analytic perspective, covariance functions are positive-definite kernels. Such kernels induce reproducing kernel Hilbert spaces (RKHS), which provide a common mathematical framework for both geostatistics and kernel-based machine learning [6, 96, 105, 121].

Under this interpretation, kriging, Gaussian process regression, support vector machines, and kernel ridge regression can all be viewed as kernel-based approximation procedures differing mainly in their assumptions concerning uncertainty, regularization, and optimization.

This equivalence is particularly clear in Gaussian process regression and smoothing-spline theory, where covariance kernels, penalty functionals, and posterior mean estimators are closely connected [96, 121].

This observation establishes an important theoretical bridge between geostatistics and machine learning.

2.5 Spatial inverse problems and regularization

Many spatial prediction problems are intrinsically ill-posed. Sparse sampling, measurement noise, and scale mismatches imply that multiple spatial fields may explain the observations equally well.

Let

$$A : \mathcal{H} \to \mathbb{R}^n \tag{15}$$

denote an observation operator and let

$$y = Af + \varepsilon \tag{16}$$

represent the observed data.

Recovering the unknown field f from observations y is generally an inverse problem [38, 70].

A common regularized formulation is

$$\min_f \left[\|Af - y\|^2 + \lambda \mathcal{R}(f) \right], \tag{17}$$

where:

- A is the observation operator;
- $\mathcal{R}(f)$ is a regularization functional;
- and λ controls smoothness or complexity.

Different regularization choices generate different classes of spatial models:

- Tikhonov regularization promotes smoothness [112];
- total variation regularization preserves discontinuities;
- Sobolev regularization imposes differentiability constraints.

These formulations are especially relevant for environmental sciences, where datasets are often incomplete, noisy, and multiscale.

2.6 Functional analysis and machine learning

Machine learning increasingly relies on functional analytic concepts. Neural networks can be interpreted as nonlinear operators between Banach spaces, while deep learning optimization occurs in high-dimensional parameterized function spaces.

More recently, neural operators and physics-informed neural networks have extended this connection further by learning mappings between functional spaces rather than finite-dimensional vectors [82].

This perspective is especially important in spatial-temporal modeling, where many environmental systems are governed by partial differential equations:

$$\mathcal{L}u = f, \tag{18}$$

where $\mathcal{L}$ is a differential operator.

Examples include groundwater flow, atmospheric transport, heat diffusion, contaminant propagation, and ocean circulation.

The integration of geostatistics, functional analysis, inverse problems, and machine learning therefore represents a natural direction for next-generation spatial modeling.

3 Geospatial Foundation Models and Learned Spatial Representations

3.1 From handcrafted features to learned representations

For several decades, spatial prediction from remote sensing data relied heavily on manually engineered features derived from physical intuition and domain expertise. Rather than learning representations directly from raw observations, analysts constructed variables believed to capture relevant environmental processes.

One of the most widely used examples is the Normalized Difference Vegetation Index (NDVI):

$$NDVI = \frac{\rho_{NIR} - \rho_R}{\rho_{NIR} + \rho_R}, \tag{19}$$

where ρ_{NIR} and ρ_R denote near-infrared and red reflectance, respectively. The index was introduced as a compact spectral proxy for vegetation condition and remains one of the most influential examples of handcrafted remote-sensing feature design [100, 114].

NDVI and related vegetation indices were designed to summarize complex spectral information into interpretable variables associated with vegetation vigor, biomass, canopy structure, and photosynthetic activity. Similar approaches generated numerous spectral indices targeting specific environmental properties such as moisture content, burn severity, chlorophyll concentration, urbanization, and water extent [46, 66, 86].

Beyond spectral indices, spatial analysis frequently incorporated texture measures derived from local neighborhoods. Examples include variance filters, gray-level co-occurrence matrices, entropy measures, and wavelet-based descriptors. These variables attempted to represent spatial organization, heterogeneity, and landscape structure using handcrafted transformations [57, 83].

From a machine learning perspective, these approaches can be interpreted as manually defining a feature map

$$\phi : X \to \mathbb{R}^p, \tag{20}$$

where X denotes the original observation space and $\phi(X)$ represents engineered predictors.

Although these methods remain useful and physically interpretable, they possess an important limitation: the representation itself is fixed a priori. The analyst determines which features are important before the learning process begins.

Recent developments in machine learning have shifted this paradigm. Instead of manually constructing features, modern representation learning attempts to learn useful representations directly from data [9]. The objective is no longer merely to estimate a prediction function

$$f(\mathbf{x}), \tag{21}$$

but also to learn an internal representation

$$\mathbf{z} = \Phi(\mathbf{x}), \tag{22}$$

where $\mathbf{z}$ captures latent structure relevant across many downstream tasks.

This transition mirrors developments in natural language processing and computer vision, where learned embeddings increasingly replaced handcrafted linguistic or visual descriptors. Earth observation is now undergoing a similar transformation [15, 37, 59].

3.2 Foundation models for Earth observation

The emergence of foundation models represents one of the most significant recent developments in Earth observation analytics. Their appearance reflects a broader transition occurring across machine learning, where models are increasingly trained on massive and heterogeneous datasets before being adapted to specific downstream tasks [10]. Rather than learning a single prediction objective, foundation models attempt to acquire general-purpose representations that capture regularities present across large collections of observations. The resulting representations can then be reused for many applications, reducing the need to design specialized models for each individual problem.

In the Earth observation domain, foundation models are typically trained using self-supervised or weakly supervised objectives on vast archives of satellite imagery and related geospatial datasets [27, 68]. The underlying idea is that the Earth system contains rich spatial, spectral, and temporal structure that can be learned directly from observations without requiring extensive manual labeling. By exploiting these large archives, the models learn internal representations that are expected to transfer across regions, sensors, environmental conditions, and analytical tasks.

Several prominent examples have emerged in recent years. AlphaEarth explores large-scale geospatial representation learning from heterogeneous Earth observation datasets, emphasizing the construction of representations that remain useful across multiple tasks and geographical contexts [14]. Prithvi follows a similar philosophy, leveraging large Earth observation archives to support applications such as land-cover mapping, environmental monitoring, disaster assessment, and climate-related analysis [68]. SatMAE extends masked autoencoder architectures to remote sensing imagery by reconstructing intentionally hidden portions of an image during training, thereby encouraging the model to learn meaningful spatial and spectral structure without relying heavily on labeled examples [27]. Clay likewise focuses on learning reusable geospatial embeddings from large collections of observations that can subsequently support a variety of downstream analyses [24].

Although these systems differ substantially in architecture, training strategy, and data sources, they share a common conceptual objective. Their primary goal is not to produce a specific environmental prediction, but rather to learn a representation space that captures useful information about the Earth system. Prediction, classification, segmentation, retrieval, anomaly detection, and change detection can then be performed on top of this learned representation. Consequently, the most important output of a foundation model is often not a map or a forecast, but a latent representation that can serve as a foundation for many subsequent tasks.

3.3 Embeddings as spatial operators

From a functional analytic perspective, foundation models can be interpreted as operators acting on spaces of Earth observations, paralleling broader work on learned operators and function-space mappings [76]. Let

$$X \tag{23}$$

denote a space that may contain multispectral imagery, radar measurements, temporal sequences, topographic variables, or other forms of geospatial information. A foundation model defines a mapping

$$\Phi : X \to \mathbb{R}^d, \tag{24}$$

where the output dimension d is typically much smaller than the dimensionality of the original observation space. The resulting vector

$$\mathbf{z} = \Phi(x) \tag{25}$$

is commonly referred to as an embedding and serves as a compact representation of the original observation.

Unlike traditional feature engineering, where the analyst explicitly specifies the transformation from observations to predictors, the mapping Φ is learned directly from data. The embedding therefore emerges from the optimization process itself rather than from prior assumptions about which variables or indices should be important. In this sense, foundation models shift the focus from manually designing representations to learning them automatically from large collections of observations.

For geospatial systems, however, observations rarely exist independently of space and time. The meaning of a measurement often depends not only on the observed signal but also on where and when it was acquired. A more realistic formulation is therefore

$$\Phi(x, \mathbf{s}, t), \tag{26}$$

where x denotes the observed signal, $\mathbf{s}$ denotes spatial location, and t denotes time. Under this interpretation, the embedding becomes a spatial-temporal representation that jointly incorporates information about the observation itself and its environmental context.

Such embeddings may implicitly encode spectral relationships, spatial patterns, temporal dynamics, land-surface structure, environmental similarity, and other latent properties that are difficult to describe through handcrafted features alone. This perspective is important because it clarifies what foundation models actually learn. Their primary objective is not to estimate environmental variables directly, but to construct operators that transform observations into useful latent representations. Prediction then becomes a secondary task performed within the learned embedding space. Foundation models should therefore be understood principally as representation-learning systems rather than prediction systems.

3.4 Spatial inference remains an open problem

The rapid success of geospatial foundation models has generated considerable enthusiasm and, in some cases, the perception that traditional spatial statistics may eventually become unnecessary. Such conclusions are difficult to justify. Foundation models unquestionably address an important component of the spatial inference problem by providing powerful mechanisms for extracting informative representations from large and heterogeneous datasets. However, representation learning and spatial inference are not identical problems, and many of the challenges discussed throughout this chapter remain largely unresolved [30, 47].

One important example concerns spatial support. An embedding extracted from a 10-meter satellite pixel, a 1-kilometer climate grid cell, or an areal average over an agricultural field does not automatically resolve the change-of-support problem. Even if the learned representations are highly informative, they may still correspond to fundamentally different observational supports. Consequently, many of the difficulties associated with comparing, fusing, or transferring information across scales remain present.

Scale dependence presents a closely related challenge. Foundation models often appear capable of learning multiscale structure implicitly, yet this does not necessarily imply a formal understanding of how variance, covariance, or uncertainty evolve across scales. The model may successfully encode information from multiple resolutions without providing an explicit statistical framework describing the relationships among them. Similar considerations apply to spatial covariance. Two observations may occupy nearby locations in an embedding space while possessing very different covariance structures in the underlying physical field. Learned similarity and statistical dependence are related concepts, but they are not equivalent.

Perhaps the most important limitation concerns uncertainty. Most current foundation models produce deterministic embeddings of the form

$$\mathbf{z} = \Phi(x, \mathbf{s}, t), \tag{27}$$

which provide no direct characterization of prediction uncertainty, spatial uncertainty, support uncertainty, or model uncertainty. Yet these quantities remain central to scientific inference and environmental decision-making. A representation may be highly informative while still offering little guidance regarding confidence, reliability, or risk.

For this reason, foundation models should not be interpreted as replacements for geostatistics, uncertainty quantification, or spatial inference theory. A more productive perspective is to view them as powerful new representation operators that can be integrated into broader spatial modeling frameworks. Within such a framework, foundation models provide learned representations, machine learning supplies flexible predictive mappings, and geostatistics contributes explicit treatment of spatial dependence and uncertainty. The future of spatial analysis will likely emerge not from the dominance of any single paradigm, but from the integration of these complementary perspectives within a unified inferential framework.

4 Spatial dependence and stochastic fields

4.1 Spatial random fields

Geostatistics models spatial phenomena as realizations of random fields defined over continuous domains [23, 30, 85].

Let

$$Z(\mathbf{s}), \quad \mathbf{s} \in D, \tag{28}$$

denote a spatial random field.

A common decomposition is

$$Z(\mathbf{s}) = m(\mathbf{s}) + \varepsilon(\mathbf{s}), \tag{29}$$

where:

- $m(\mathbf{s})$ represents the deterministic trend;
- $\varepsilon(\mathbf{s})$ denotes a zero-mean spatially correlated residual process.

This decomposition separates large-scale deterministic variation from local stochastic dependence [36, 52, 85].

4.2 Spatial autocorrelation

The defining property of spatial data is spatial autocorrelation: nearby observations tend to be more similar than distant observations [25, 89].

This principle is often summarized through Tobler's first law of geography:

> "Everything is related to everything else, but near things are more related than distant things."

[113].

Spatial autocorrelation implies that observations cannot generally be treated as independent samples [5, 30].

Ignoring this dependence can lead to:

- biased uncertainty estimates;
- inflated validation scores;
- unstable parameter inference;
- and misleading predictive maps.

4.3 Covariance and semivariogram functions

The covariance structure of a spatial field [30, 108] is represented by

$$C(\mathbf{h}) = \mathrm{Cov}\left[Z(\mathbf{s}), Z(\mathbf{s}+\mathbf{h})\right]. \tag{30}$$

Under intrinsic stationarity, spatial dependence is equivalently represented through the semivariogram [84, 85]:

$$\gamma(\mathbf{h}) = \frac{1}{2}\mathrm{Var}\left[Z(\mathbf{s}) - Z(\mathbf{s}+\mathbf{h})\right]. \tag{31}$$

The semivariogram quantifies how dissimilarity evolves with spatial separation vector $\mathbf{h}$ [52, 122].

Three quantities are commonly interpreted:

- the **nugget**, associated with measurement error or microscale variability;
- the **sill**, corresponding to total variance;
- the **range**, beyond which spatial dependence becomes negligible.

Variogram analysis remains one of the central tools of geostatistics because it provides direct information about scale-dependent spatial organization [23, 122].

4.4 Stationarity and anisotropy

Most classical geostatistical methods rely on stationarity assumptions.

Second-order stationarity requires:

$$\mathbb{E}[Z(\mathbf{s})] = \mu, \tag{32}$$

and

$$\mathrm{Cov}\left[Z(\mathbf{s}), Z(\mathbf{s}+\mathbf{h})\right] = C(\mathbf{h}), \tag{33}$$

meaning that covariance depends only on spatial separation. These assumptions provide the mathematical conditions under which covariance and variogram models can be estimated and used for spatial prediction [23, 30].

In practice, many environmental processes violate strict stationarity because of trends, discontinuities, or heterogeneous physical controls [8, 103].

Anisotropy introduces additional complexity because spatial dependence may vary by direction rather than distance alone [23, 52].

These properties are not merely diagnostic details. They determine whether a spatial covariance model is mathematically defensible.

4.5 Scale dependence and multiscale variability

Spatial dependence is inherently multiscale. Environmental systems often exhibit:

- short-range local variability;
- intermediate-scale structures;
- and large-scale deterministic gradients.

A single covariance structure may therefore fail to represent all relevant spatial scales adequately [30, 120].

This motivates multiscale approaches such as nested variograms, wavelet decompositions, and factorial kriging [50, 53, 120].

Understanding scale dependence is fundamental because predictive performance, uncertainty quantification, and interpretability all depend strongly on the scale at which spatial organization is modeled.

5 Geostatistical modeling

5.1 From covariance structure to spatial prediction

Geostatistics uses the covariance structure of spatial fields to construct optimal linear predictors at unsampled locations [30, 85].

Let observations be available at locations

$$\{\mathbf{s}_1, \ldots, \mathbf{s}_n\}, \tag{34}$$

and let

$$Z(\mathbf{s}_0) \tag{35}$$

denote the unknown value at target location $\mathbf{s}_0$.

Ordinary kriging estimates the target value as

$$\hat{Z}(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i), \tag{36}$$

where the weights λ_i are determined by minimizing the prediction variance subject to unbiasedness constraints.

The kriging system is obtained from the covariance structure of the field and yields the best linear unbiased estimator (BLUE) under second-order assumptions.

Unlike deterministic interpolation procedures, kriging explicitly quantifies prediction uncertainty through the kriging variance:

$$\sigma_K^2(\mathbf{s}_0) = \mathrm{Var}\left[Z(\mathbf{s}_0) - \hat{Z}(\mathbf{s}_0)\right]. \tag{37}$$

This uncertainty estimate is fundamental because two predictions with identical expected values may possess very different confidence levels depending on local sampling density and covariance structure.

5.2 Ordinary, universal, and regression kriging

Different kriging variants correspond to different assumptions concerning the mean structure of the spatial field.

Ordinary kriging assumes an unknown but locally constant mean:

$$Z(\mathbf{s}) = \mu + \varepsilon(\mathbf{s}). \tag{38}$$

Universal kriging incorporates deterministic spatial trends:

$$Z(\mathbf{s}) = m(\mathbf{s}) + \varepsilon(\mathbf{s}), \tag{39}$$

where

$$m(\mathbf{s}) = \sum_{k=1}^{p} \beta_k f_k(\mathbf{s}). \tag{40}$$

Regression kriging combines regression models with geostatistical modeling of residual spatial dependence [61, 62]:

$$Z(\mathbf{s}) = f(\mathbf{x}(\mathbf{s})) + \varepsilon(\mathbf{s}), \tag{41}$$

where:

- $f(\mathbf{x}(\mathbf{s}))$ captures deterministic covariate effects;
- $\varepsilon(\mathbf{s})$ captures remaining spatial autocorrelation.

This formulation already suggests a bridge between geostatistics and machine learning.

5.3 Sequential Gaussian simulation

Kriging produces smooth optimal estimates but tends to underestimate local variability because predictions correspond to conditional means.

In many environmental and geological applications, reproducing the full spatial variability is more important than obtaining a single minimum-variance estimate.

Sequential Gaussian Simulation (SGS) addresses this problem by generating conditional realizations of the spatial field [23, 34].

The method assumes that the transformed variable follows a Gaussian distribution after normal-score transformation:

$$Y(\mathbf{s}) = \phi\left(Z(\mathbf{s})\right), \tag{42}$$

where $\phi(\cdot)$ denotes the Gaussian anamorphosis transform.

Simulation then proceeds sequentially over the domain:

1. a random path through all unsampled locations is generated;
2. simple kriging is performed at the current location;
3. a value is randomly drawn from the conditional Gaussian distribution:

$$Y(\mathbf{s}_0) \sim \mathcal{N}\left(\hat{Y}(\mathbf{s}_0), \sigma_K^2(\mathbf{s}_0)\right); \tag{43}$$

4. the simulated value is added to the conditioning dataset;
5. the process continues until all locations are simulated.

Unlike kriging, SGS preserves:

- the histogram of the variable;
- the spatial covariance structure;

- realistic local variability;
- spatial uncertainty patterns.

Multiple realizations can be generated:

$$\{Z^{(1)}(\mathbf{s}), \ldots, Z^{(m)}(\mathbf{s})\}, \tag{44}$$

allowing uncertainty propagation into downstream environmental or decision models.

Pointwise uncertainty can then be estimated empirically:

$$\operatorname{Var}[Z(\mathbf{s})] \approx \frac{1}{m-1} \sum_{k=1}^{m} \left(Z^{(k)}(\mathbf{s}) - \bar{Z}(\mathbf{s})\right)^2. \tag{45}$$

Sequential Gaussian simulation is widely used in hydrogeology, reservoir characterization, mining, and climate applications where extreme values and spatial heterogeneity are critical.

5.4 Sequential indicator simulation and non-Gaussian methods

Many environmental variables are non-Gaussian, multimodal, or strongly bounded.

Indicator approaches avoid strict Gaussian assumptions by transforming the variable into threshold-based indicators:

$$I(\mathbf{s}; z_c) = \begin{cases} 1, & Z(\mathbf{s}) \le z_c, \\ 0, & Z(\mathbf{s}) > z_c. \end{cases} \tag{46}$$

Indicator kriging estimates conditional exceedance probabilities:

$$P\left(Z(\mathbf{s}) \le z_c \mid \mathcal{D}\right), \tag{47}$$

where $\mathcal{D}$ denotes the observed dataset.

Sequential Indicator Simulation (SIS) extends this framework to generate conditional realizations without relying on Gaussianity [52].

At each location, values are sampled from locally estimated conditional cumulative distributions reconstructed from multiple indicator thresholds.

These approaches are particularly useful when:

- distributions are highly skewed;
- sharp boundaries exist;
- threshold exceedances dominate the problem;
- extreme events are physically important.

Other non-Gaussian simulation methods include:

- plurigaussian simulation;
- multiple-point geostatistics;
- truncated Gaussian models;
- copula-based spatial simulation.

Multiple-point geostatistics is especially important because it captures complex spatial patterns that cannot be represented through two-point covariance functions alone.

5.5 Cokriging and multivariate spatial dependence

Many environmental variables are physically related across space. Geostatistics can incorporate these relationships through multivariate covariance modeling.

Let

$$Z_1(\mathbf{s}), \ldots, Z_p(\mathbf{s}) \tag{48}$$

represent multiple spatial variables.

Cokriging exploits both direct and cross-covariance structures:

$$C_{ij}(\mathbf{h}) = \mathrm{Cov}\left[Z_i(\mathbf{s}), Z_j(\mathbf{s}+\mathbf{h})\right]. \tag{49}$$

The linear model of coregionalization represents these covariance functions as combinations of elementary spatial structures [52, 120].

This framework is especially useful when:

- the target variable is sparsely sampled;
- secondary variables are densely available;
- or multiple sensors observe related physical processes.

5.6 Block kriging and support effects

Spatial observations possess support. A point sample, a raster pixel, and a regional average are not equivalent quantities.

If support differences are ignored, prediction uncertainty may become artificially optimistic and spatial interpretation may become invalid [31, 97].

Block kriging addresses this problem by estimating averages over finite spatial regions rather than point values.

Let

$$B \subset D \tag{50}$$

denote a spatial block. The block prediction becomes:

$$\hat{Z}(B) = \frac{1}{|B|} \int_B \hat{Z}(\mathbf{s})\, d\mathbf{s}. \tag{51}$$

Because averaging reduces local variability, block kriging typically produces lower uncertainty than point kriging.

This distinction is particularly important in environmental applications where management decisions operate over finite spatial units rather than mathematical points.

5.7 Indicator kriging and uncertainty modeling

Classical kriging methods often rely implicitly on Gaussian assumptions.

However, many environmental variables are strongly skewed, heteroscedastic, or heavy-tailed.

Indicator kriging addresses this issue by transforming the variable into binary indicators:

$$I(\mathbf{s}; z_c) = \begin{cases} 1, & Z(\mathbf{s}) \leq z_c, \\ 0, & Z(\mathbf{s}) > z_c, \end{cases} \tag{52}$$

where z_c is a threshold value.

The resulting predictions correspond to exceedance probabilities rather than point estimates [52].

This probabilistic interpretation is often more useful for risk analysis and environmental decision support than a single interpolated surface.

5.8 Spatiotemporal geostatistics

Many environmental systems evolve continuously in both space and time.

Spatiotemporal geostatistics extends covariance modeling to joint space-time dependence:

$$C(\mathbf{h}, u) = \mathrm{Cov}\left[Z(\mathbf{s}, t), Z(\mathbf{s} + \mathbf{h}, t + u)\right], \tag{53}$$

where $\mathbf{h}$ denotes spatial lag and u temporal lag.

Separable covariance models assume:

$$C(\mathbf{h}, u) = C_s(\mathbf{h}) C_t(u), \tag{54}$$

although nonseparable formulations are often more realistic for dynamic environmental processes.

Spatiotemporal kriging and simulation are increasingly important in:

- meteorology;
- oceanography;
- hydrology;
- climate reanalysis;
- air quality modeling.

5.9 Strengths and limitations of geostatistics

Geostatistics is particularly strong when:

- spatial autocorrelation is dominant;
- uncertainty quantification is essential;
- sampling density is limited;
- interpretability of spatial structure matters.

Its mathematical strength derives from the explicit modeling of covariance structure and stochastic dependence.

However, classical geostatistics also possesses important limitations.

Most methods rely strongly on second-order stationarity or intrinsic stationarity assumptions [23, 85].

Under strong heterogeneity, abrupt discontinuities, or highly nonlinear covariate interactions, variogram estimation may become unstable or physically ambiguous.

In addition, covariance matrix inversion introduces computational limitations for very large datasets because standard kriging scales approximately as

$$\mathcal{O}(n^3). \tag{55}$$

Consequently, geostatistics alone may become insufficient when the dominant signal arises primarily from high-dimensional nonlinear predictor interactions rather than spatial covariance itself.

6 Machine learning for spatial prediction

6.1 Machine learning as nonlinear operator approximation

Machine learning methods approach spatial prediction differently from classical geostatistics [58, 71]. Rather than modeling covariance structures explicitly, most machine learning algorithms attempt to learn nonlinear mappings between predictors and responses [58, 116]:

$$y_i = f(\mathbf{x}_i) + \eta_i, \tag{56}$$

where:

- $\mathbf{x}_i$ denotes predictor variables;
- f is an unknown nonlinear operator;
- η_i represents residual error.

From a functional analytic perspective, machine learning can therefore be interpreted as nonlinear operator approximation in high-dimensional feature spaces [51].

In spatial applications, predictor variables may include:

- environmental covariates;
- topographic attributes;
- remote sensing variables;
- climatic indicators;
- spatial coordinates.

Importantly, including coordinates as predictors is not equivalent to explicitly modeling spatial covariance structure [63, 88].

A model may exploit location information while still failing to represent stochastic spatial dependence probabilistically [30, 98].

6.2 Kernel methods and Gaussian processes

Several machine learning methods possess strong mathematical connections with geostatistics [96, 105].

Kernel methods define similarity through positive-definite kernels [105]:

$$K(\mathbf{x}_i, \mathbf{x}_j). \tag{57}$$

These kernels induce reproducing kernel Hilbert spaces and allow high-dimensional nonlinear approximation through implicit feature mappings [48, 105].

Gaussian process regression is particularly close to kriging because it models functions probabilistically through covariance kernels:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), K(\mathbf{x}, \mathbf{x}')). \tag{58}$$

Under suitable assumptions, Gaussian process regression and kriging become mathematically equivalent up to differences in terminology, prior interpretation, and computational implementation [96, 105, 108].

6.3 Tree-based and ensemble methods

Tree-based methods are widely used in spatial prediction because they handle nonlinear interactions, heterogeneous predictors, and missing data effectively [11, 12, 63].

Random forests aggregate multiple decision trees:

$$\hat{f}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}), \tag{59}$$

where T_b denotes the b-th tree [11].

Gradient boosting methods sequentially minimize residual prediction error through additive weak learners [22, 44].

These methods often outperform classical geostatistical approaches when:

- many predictors are available;
- relationships are strongly nonlinear;
- interactions are complex;
- or spatial autocorrelation is secondary relative to environmental controls.

6.4 Deep learning and spatial-temporal representation

Deep learning extends machine learning to hierarchical nonlinear representation learning [51, 80].

Convolutional neural networks are particularly useful for raster and image-based spatial prediction because they exploit local spatial patterns directly [78].

Recurrent and transformer architectures additionally support spatial-temporal modeling [117].

More recently, graph neural networks and neural operators have generalized these ideas toward irregular spatial domains and operator learning [72, 81, 82].

These developments are especially relevant for:

- climate modeling;
- remote sensing;
- oceanography;
- atmospheric transport;
- environmental forecasting.

6.5 Validation under spatial dependence

A major challenge in spatial machine learning is validation bias [88, 98].

Most machine learning theory assumes that training and testing samples are approximately independent [58].

Spatial autocorrelation violates this assumption because nearby observations are statistically related [30, 89].

Consequently, random cross-validation often produces overly optimistic performance estimates [98, 115].

Spatially blocked validation reduces this bias by separating training and testing regions geographically [92, 115].

This distinction is fundamental because a model may appear highly accurate under random validation while failing to generalize to truly unobserved spatial regions.

6.6 Strengths and limitations of machine learning

Machine learning methods are particularly strong when:

- predictor dimensionality is large;
- nonlinear interactions dominate;
- large datasets are available;
- predictive accuracy is prioritized.

Their flexibility allows the representation of highly complex relationships that are difficult to capture through classical covariance models [58, 71].

However, important limitations remain.

Most conventional machine learning algorithms do not explicitly model spatial covariance structure in the probabilistic sense used in geostatistics [30, 52].

As a result:

- uncertainty quantification may be weak;
- spatial dependence may remain hidden in residuals;
- extrapolation may become unstable;
- and validation procedures may become biased.

In environmental sciences, these limitations are especially important because uncertainty propagation and spatial interpretability are often as important as predictive accuracy itself.

7 Spatial uncertainty and inverse problems

7.1 Prediction versus uncertainty

Spatial prediction alone is insufficient for scientific inference [30, 52].

A map containing only expected values may appear visually convincing while concealing large uncertainty arising from sparse sampling, measurement noise, or model instability [23, 36].

Consequently, spatial analysis should be interpreted as the joint problem of:

- estimating the spatial field;
- quantifying uncertainty;
- and characterizing the stability of the inference process.

This distinction becomes critical in environmental applications where sampling is irregular and observations are expensive or incomplete [8, 30].

7.2 Spatial prediction as an inverse problem

Spatial estimation can be interpreted mathematically as an inverse problem [38, 70, 111]. Let

$$A : \mathcal{H} \to \mathbb{R}^n \tag{60}$$

denote an observation operator mapping an unknown spatial field f into observations:

$$y = Af + \varepsilon. \tag{61}$$

The inverse problem consists of recovering f from y [7].

This problem is generally ill-posed because:

- observations are finite;
- measurements contain noise;
- spatial support may vary;
- multiple fields may explain the same observations.

Regularization is therefore essential [38, 112].

A common formulation is:

$$\min_f \left[\|Af - y\|^2 + \lambda \mathcal{R}(f) \right], \tag{62}$$

where:

- $\mathcal{R}(f)$ controls smoothness or complexity;
- λ balances fidelity and regularization.

This formulation connects geostatistics, variational methods, Bayesian inference, and machine learning through a common mathematical framework [51, 70, 96].

7.3 Regularization and spatial smoothness

Different regularization choices impose different assumptions concerning the geometry of spatial fields.

Tikhonov regularization penalizes large norms [38, 112]:

$$\mathcal{R}(f) = \|f\|^2. \tag{63}$$

Sobolev regularization incorporates derivatives [1, 13]:

$$\mathcal{R}(f) = \|\nabla f\|^2. \tag{64}$$

Total variation regularization instead preserves discontinuities [101]:

$$\mathcal{R}(f) = \int_D |\nabla f(\mathbf{s})| \, d\mathbf{s}. \tag{65}$$

These choices are not merely computational details. They encode assumptions concerning smoothness, continuity, and spatial structure [38, 70].

7.4 Uncertainty propagation in geostatistics

One major strength of geostatistics is explicit uncertainty representation [30, 52].

Kriging variance quantifies local prediction uncertainty [30, 85]:

$$\sigma_K^2(\mathbf{s}) = \mathrm{Var}\left[Z(\mathbf{s}) - \hat{Z}(\mathbf{s})\right]. \tag{66}$$

Importantly, this uncertainty depends not only on local sampling density but also on the covariance structure of the field [108].

Conditional simulation extends this framework by generating multiple equiprobable spatial realizations consistent with observations and covariance structure [23, 35].

This allows:

- uncertainty propagation;
- risk assessment;
- probability estimation;
- spatial decision analysis.

7.5 Uncertainty in machine learning

Uncertainty representation remains more difficult in many machine learning frameworks [47].

Deterministic algorithms often provide point predictions without probabilistic interpretation [51].

Several approaches attempt to address this limitation:

- Bayesian neural networks [45];
- ensemble learning [79];
- quantile regression [73];
- conformal prediction [4, 119];
- Gaussian process models [96].

However, uncertainty estimates from machine learning models may still fail to represent spatial covariance structure explicitly [47, 98].

This distinction is important because predictive uncertainty and spatial dependence are not identical concepts [30].

7.6 Sampling density and spatial reliability

The reliability of spatial inference depends strongly on sampling configuration [30, 122].

Sparse or clustered sampling may fail to capture the true covariance structure of the field [29, 36].

Under such conditions:

- variogram estimation becomes unstable;
- interpolation artifacts may arise;
- uncertainty estimates may become misleading.

This problem is especially severe in heterogeneous environments where multiple spatial scales coexist [23, 120].

Consequently, visually smooth maps should never be interpreted as evidence of reliable spatial inference by themselves [52].

7.7 Transformation and variance stabilization

Environmental variables are frequently skewed, heteroscedastic, and heavy-tailed [23, 52].

Because variograms depend on squared differences, extreme skewness may destabilize covariance estimation [30].

Transformations therefore play an important role in spatial modeling [35, 52].

Gaussian anamorphosis seeks a transformation [23, 35]:

$$Y(\mathbf{s}) = \phi(Z(\mathbf{s})) \tag{67}$$

such that $Y(\mathbf{s})$ becomes approximately Gaussian.

Spatial modeling is then performed in transformed space before back-transformation [69].

These procedures are not cosmetic statistical adjustments. They are often necessary for stable covariance modeling and reliable uncertainty estimation [35, 52].

8 Support theory and multiscale spatial representation

8.1 Spatial support as part of the data definition

Spatial observations are not defined solely by their values and locations. They are also defined by their spatial support [30, 54].

Support refers to the spatial region over which a measurement is effectively averaged, integrated, or observed [30, 97].

A point sample, a raster pixel, a satellite footprint, and an administrative average are fundamentally different spatial objects, even when they refer to the same environmental variable [31, 49].

Let

$$B \subset D \tag{68}$$

denote a spatial support region. The observed quantity may then be represented as:

$$Z(B) = \frac{1}{|B|} \int_B Z(\mathbf{s})\, d\mathbf{s}. \tag{69}$$

This formulation emphasizes that many observations are not exact point values, but spatial averages over finite domains [54, 97].

The importance of support is often underestimated in practical analysis. However, support directly affects:

- **Variance structure:** averaging over larger supports reduces local variability and smooths short-range fluctuations.
- **Covariance behavior:** spatial dependence changes with support because smoothing modifies correlation structure.
- **Interpretability:** values observed at different supports do not necessarily represent the same physical quantity.
- **Prediction uncertainty:** uncertainty estimated at point scale cannot automatically be transferred to block-scale prediction.

Ignoring support differences can generate:

- **Biased spatial comparisons:** point observations may be incorrectly compared with areal averages or coarse pixels.
- **Artificial smoothing:** support mismatch can create the illusion of spatial continuity where none exists physically.
- **Overconfident uncertainty estimates:** uncertainty may be underestimated when averaging effects are ignored.

- **Inconsistent multiscale predictions:** models calibrated at one support may fail when transferred to another spatial resolution.

Consequently, support should be interpreted as a fundamental component of the measurement process rather than a secondary technical detail [30, 31].

8.2 The change-of-support problem

A fundamental consideration in spatial data analysis is the concept of *support*—the spatial, temporal, or spatiotemporal unit over which a variable is measured or aggregated [49]. This concept becomes particularly important as data are collected from multiple sources (e.g., drone imagery, proximal sensors, soil sampling, and satellite observations) at vastly different spatial resolutions [30]. When information from these sources is combined or compared, the differences in support can create misleading conclusions if not appropriately accounted for [17].

This issue, known as the **change of support problem (COSP)**, arises when spatial variables or predictions are represented at one resolution or scale, but are modeled or interpreted at another [31, 54]. For instance, high-resolution weed maps generated from UAV imagery (e.g., at 5 cm/pixel) may be aggregated to the scale of agricultural management zones (e.g., 10×10 meters), while soil nutrient data may only be available at the field level or as coarse interpolated maps from sparse sampling points [107]. When such datasets are fused without addressing the inherent difference in support, the resulting spatial estimates—whether of weed density, soil fertility, or treatment efficiency—can suffer from bias and unquantified uncertainty [106].

In spatial analysis, techniques such as kriging, block kriging, spatial aggregation, or hierarchical Bayesian modeling can be used to partially address the change of support problem [19]. Still, these methods come with their own assumptions and limitations [17]. The choice of technique, resolution, and representation directly influences the derived spatial patterns, model outputs, and ultimately the agronomic decisions made. Without careful handling, COSP can result in unjustified confidence in high-resolution maps that are not statistically or biologically defensible [2].

This issue is pervasive in environmental sciences because most modern datasets are intrinsically multiscale [23, 120].

Examples include:

- **Relating point soil samples to satellite pixels:** field observations represent local measurements, while remote sensing products represent spatial averages over larger footprints.
- **Combining station observations with climate grids:** weather stations provide point-scale information, whereas climate products often represent coarse regional averages.
- **Aggregating local measurements into management zones:** operational decisions are frequently made over finite spatial regions rather than individual points.
- **Integrating heterogeneous remote sensing products:** different sensors may observe the same variable at incompatible spatial resolutions and supports.

Mathematically, changing support modifies both variance and covariance structure [30, 54].

Averaging over larger supports typically reduces variance [97]:

$$\mathrm{Var}[Z(B)] < \mathrm{Var}[Z(\mathbf{s})]. \tag{70}$$

At the same time, covariance structures generally become smoother and longer-ranged because local fluctuations are partially averaged out [31, 52].

These effects are not merely statistical artifacts. They reflect the fact that different supports observe different manifestations of the same latent spatial field [8].

Consequently:

- covariance models estimated at one support cannot automatically be transferred to another support;
- variograms derived from coarse-resolution products may fail to represent microscale variability;
- uncertainty estimates may become misleading if support effects are ignored.

The change-of-support problem is therefore one of the central reasons why spatial prediction cannot be reduced to ordinary interpolation [30, 54].

8.3 Block support and operational prediction

Many environmental decisions operate over finite regions rather than mathematical points [30, 52].

Agricultural management zones, ecological habitats, watersheds, administrative regions, and climate grid cells are all examples of block support [54].

In such contexts, point prediction may possess limited operational meaning because decisions are implemented over spatial aggregates.

Block kriging addresses this issue by estimating spatial averages [52, 97]:

$$\hat{Z}(B) = \frac{1}{|B|} \int_B \hat{Z}(\mathbf{s})\, d\mathbf{s}. \tag{71}$$

This distinction is important because block prediction and point prediction possess different statistical properties [30].

Block prediction generally exhibits:

- **Reduced variance:** averaging suppresses local stochastic variability.
- **Greater stability:** predictions become less sensitive to isolated observations or local anomalies.
- **Improved operational relevance:** predictions align more closely with the spatial scale of management or policy decisions.

- **Different uncertainty behavior:** block uncertainty reflects aggregated variability rather than pointwise fluctuation.

This distinction is operationally important because visually precise point maps may suggest unrealistic levels of local certainty [52].

In many environmental applications, the relevant inference problem is not:

$$Z(\mathbf{s}), \tag{72}$$

but rather:

$$Z(B). \tag{73}$$

Support-aware prediction is therefore essential for scientifically defensible decision-making [8, 54].

8.4 Scale dependence and spatial organization

Environmental systems are intrinsically multiscale [23, 120].

Observed spatial variability often reflects the superposition of:

- **Local stochastic variability:** short-range heterogeneity associated with microscale processes or measurement noise.
- **Intermediate-scale environmental gradients:** variability associated with terrain, hydrology, vegetation, or land use.
- **Large-scale deterministic forcing:** broad spatial structure generated by climate, geology, or regional circulation patterns.

A single covariance structure may therefore fail to represent all relevant spatial scales adequately [52, 120].

Nested variogram models address this issue by representing covariance through multiple spatial structures [53, 120]:

$$\gamma(\mathbf{h}) = \sum_{k=1}^{K} \gamma_k(\mathbf{h}), \tag{74}$$

where each component corresponds to a distinct spatial scale.

This decomposition is important because different physical mechanisms often dominate at different scales [120].

For example:

- microscale variability may reflect local soil heterogeneity or unresolved turbulence;
- intermediate scales may reflect topographic controls or ecological organization;
- large scales may reflect climatic forcing or geological structure.

Scale dependence strongly influences:

- predictive performance;
- uncertainty quantification;
- physical interpretability;
- transferability across regions.

A model that performs well at one scale may fail at another because the dominant physical processes change with scale.

Understanding multiscale organization is therefore central to both scientific interpretation and predictive modeling [23, 52].

8.5 Wavelets and multiresolution analysis

Functional analysis provides additional tools for multiscale spatial representation through wavelets and multiresolution decompositions [33, 83].

Wavelet expansions represent spatial fields as:

$$f(\mathbf{s}) = \sum_{j,k} c_{jk}\psi_{jk}(\mathbf{s}), \tag{75}$$

where:

- ψ_{jk} are localized basis functions representing spatial patterns;
- j controls scale or resolution;
- k controls spatial location.

Unlike Fourier methods, wavelets preserve both:

- **Spatial localization:** local features remain spatially identifiable;
- **Scale information:** different resolutions can be analyzed simultaneously.

This property is especially useful for:

- **Nonstationary spatial fields:** local statistical properties may vary across the domain.
- **Abrupt discontinuities:** sharp transitions are difficult to represent with globally smooth basis functions.
- **Multiscale environmental systems:** variability may emerge from processes acting at very different ranges.
- **Spatial-temporal variability:** localized transient structures can be represented efficiently.

Wavelet methods additionally provide:

- data compression;
- noise filtering;
- multiscale decomposition;
- sparse functional representation.

These methods establish important connections between spatial statistics, harmonic analysis, signal processing, and modern machine learning representations [83, 90].

8.6 Functional representations and spatial complexity

Functional representations become increasingly important as spatial datasets grow in dimensionality and complexity [64, 94].

Environmental datasets frequently involve:

- **Hyperspectral signatures:** observations consist of high-dimensional spectral curves rather than scalar values.
- **Temporal trajectories:** variables evolve continuously through time as spatial-temporal functions.
- **Multiscale raster stacks:** information is distributed across many spatial resolutions and covariate layers.
- **Spatial-temporal fields:** environmental systems evolve jointly across space and time.
- **Coupled physical processes:** interacting systems generate complex multivariate dependence structures.

In these situations, observations are often more naturally interpreted as functions or operators rather than finite-dimensional vectors [64, 76].

Functional data analysis extends classical statistics by representing observations in infinite-dimensional spaces [64, 94, 95].

This framework supports:

- **Functional regression:** relationships are modeled between curves, surfaces, or fields rather than scalar variables.
- **Operator learning:** mappings between functional spaces can be approximated directly.
- **Spatial-temporal decomposition:** complex variability can be separated across scales and dimensions.

- **Multiscale regularization:** smoothness and complexity constraints can be imposed across multiple resolutions simultaneously.

These ideas are increasingly relevant because environmental systems are heterogeneous, multiscale, partially observed, and dynamically coupled [30, 120].

From this perspective, spatial representation becomes not merely a problem of storing coordinates and values, but a broader problem of describing complex spatial structure within infinite-dimensional functional spaces [64, 102].

9 Data fusion and multivariate spatial analysis

9.1 Why spatial data fusion is necessary

Environmental systems are rarely observed through a single variable or sensor [32, 56].

Different observation systems provide complementary information at different resolutions, supports, accuracies, temporal frequencies, and spatial extents [21, 32].

Examples include:

- **Satellite imagery:** provides large spatial coverage and repeated observations, but often with indirect measurements, atmospheric contamination, or coarse support.
- **Field measurements:** offer direct and usually more accurate local observations, but are sparse, expensive, and irregularly distributed.
- **Climate reanalysis products:** combine physical models and observations to generate spatially complete fields, although they may contain systematic biases inherited from model assumptions.
- **Geophysical surveys:** provide indirect information about subsurface or environmental structure through electromagnetic, seismic, or radiometric responses.
- **Topographic variables:** encode terrain-driven controls such as slope, curvature, drainage, and exposure, which strongly influence many environmental processes.
- **Numerical simulation outputs:** represent physically constrained system behavior but may differ from observations because of parameter uncertainty or model simplification.

No single dataset fully characterizes the spatial process [56, 99].

Data fusion therefore seeks to combine heterogeneous information sources into a coherent spatial representation [3, 18, 56].

This problem is mathematically challenging because the datasets may differ in:

- **Support:** measurements may represent points, pixels, blocks, or integrated volumes, which are not directly comparable.

- **Scale:** spatial resolution and effective averaging scale may differ substantially across sensors.
- **Uncertainty:** different observation systems possess distinct noise structures, calibration errors, and reliability levels.
- **Sampling density:** some variables may be densely available while others are sparse or irregular.
- **Physical interpretation:** sensors may observe different manifestations of the same underlying process rather than the process itself.

Consequently, data fusion should not be interpreted merely as stacking datasets together. It is fundamentally the problem of integrating multiple incomplete observations of a latent spatial field [8, 30].

An ideal method for addressing COSP in data fusion [19] should: (i) explicitly account for differences in spatial support; (ii) allow both upscaling and downscaling of data; (iii) incorporate measurement uncertainties in predictions; (iv) integrate diverse covariates; (v) preserve consistency across scales (ensuring mass balance in upscaling and pycnophylactic properties in downscaling); and (vi) be practical for implementation in a GIS environment for various support transformations.

Addressing COSP effectively is crucial for reliable data fusion, particularly when predictions are needed at scales different from those of the original measurements. However, finding a universal statistical method that fully meets the complex requirements of COSP remains a challenging task. Castrignanò and Buttafuoco [19] provide a comprehensive review of the current methods used to address this problem, each with its own strengths and limitations. For this discussion, I arise three methods. Table 1 compares examples of statistical approaches to data fusion and the COSP. The following subsections briefly describe these statistical approaches that have been developed to tackle these issues.

9.2 Multivariate spatial dependence

Let

$$Z_1(\mathbf{s}), \ldots, Z_p(\mathbf{s}) \tag{76}$$

denote multiple spatial variables [52, 120].

Multivariate spatial analysis seeks to model both:

- **Direct spatial dependence:** the autocorrelation structure of each individual variable across space.
- **Cross-dependence between variables:** statistical relationships linking different variables through shared spatial organization or common physical controls.

Cross-covariance functions are defined as:

$$C_{ij}(\mathbf{h}) = \mathrm{Cov}\left[Z_i(\mathbf{s}), Z_j(\mathbf{s}+\mathbf{h})\right]. \tag{77}$$

Table 1: Comparison of Statistical Approaches to Data Fusion and Change of Support Problem

Method	***Support Handling***	***Scale Direction***	***Uncertainty Treatment***	***GIS Integration Capability***
Geographically Weighted Regression (GWR)	Implicitly through spatial weighting kernels	Primarily downscaling	Limited; error estimates local but not probabilistic	Good, point-to-point or point-to-area via local models
Multiscale Spatial Tree Models	Hierarchical partitions support multi-resolution inference	Both upscaling and downscaling	Moderate; requires post hoc uncertainty estimation	Moderate; integration requires custom implementation
Bayesian Hierarchical Models	Explicit through latent spatial process modeling	Both upscaling and downscaling	Strong; uncertainty is fully propagated	Moderate to high; implementation via external tools like R-INLA or Stan
Multivariate Geostatistics (e.g., cokriging)	Explicit using variogram models and kriging equations	Both; area-to-point and block predictions supported	Moderate to strong; kriging variance available	Good; well-supported in GIS geostatistics toolkits

These structures allow information from secondary variables to improve prediction of sparsely sampled targets.

This principle is especially valuable when:

- **Direct sampling is expensive:** some environmental variables require costly laboratory analysis or difficult field acquisition.
- **Auxiliary variables are densely available:** remote sensing products or terrain attributes may provide nearly continuous spatial coverage.
- **Variables share common physical controls:** correlated processes may reveal latent environmental mechanisms that are not directly observable through a single variable alone.

For example, soil moisture, vegetation indices, topography, and temperature may all exhibit coupled spatial structure because they respond jointly to hydrological and climatic forcing.

Multivariate modeling therefore allows the analyst to exploit both within-variable continuity and between-variable relationships [40, 120].

9.3 The linear model of coregionalization

A classical framework for multivariate geostatistics is the linear model of coregionalization (LMC) [120].

The LMC represents covariance structures as combinations of shared elementary spatial components:

$$C_{ij}(\mathbf{h}) = \sum_{k=1}^{K} b_{ij}^{(k)} g_k(\mathbf{h}), \tag{78}$$

where:

- $g_k(\mathbf{h})$ are elementary covariance structures associated with specific spatial scales or spatial processes;
- $b_{ij}^{(k)}$ are coregionalization coefficients controlling how strongly each variable participates in each spatial structure.

This decomposition permits:

- **Multiscale covariance modeling:** different spatial ranges can be represented simultaneously within the same framework.
- **Cross-variable interpolation:** information from correlated variables can improve prediction where direct observations are sparse.
- **Spatial process decomposition:** distinct physical mechanisms may be separated according to their characteristic spatial scales.

The LMC is particularly important because environmental systems rarely operate through a single spatial mechanism. Large-scale climate forcing, intermediate-scale terrain effects, and local stochastic heterogeneity may coexist simultaneously.

The framework therefore provides both predictive capability and physical interpretability [120].

9.4 Factorial kriging and scale decomposition

Factorial kriging extends the LMC by decomposing total spatial variability into approximately independent spatial components [53, 120].

A spatial field may then be represented as:

$$Z(\mathbf{s}) = \sum_{k=1}^{K} Y_k(\mathbf{s}), \tag{79}$$

where each component $Y_k(\mathbf{s})$ corresponds to a distinct spatial scale.

This decomposition supports:

- **Separation of short-range and long-range variability:** local noise can be distinguished from broad regional structure.
- **Identification of dominant spatial scales:** different physical mechanisms may dominate at different ranges.
- **Interpretation of physical driving mechanisms:** scale decomposition often reveals meaningful environmental organization.
- **Multiscale environmental analysis:** different scales can be analyzed independently depending on the scientific question.

For example:

- microscale variability may reflect local heterogeneity or measurement noise;
- intermediate scales may reflect geomorphology or land-use patterns;
- large scales may reflect climatic or geological forcing.

Factorial kriging is therefore useful not only for prediction, but also for scientific interpretation and process understanding [53, 120].

9.5 Geographically Weighted Regression

Geographically Weighted Regression (GWR) is a local regression technique that allows model coefficients to vary across geographic space [16, 42]. This flexibility enables the model to capture non-stationary relationships between a dependent variable and its predictors [26, 55, 87], which is useful in precision agriculture where local field conditions often lead to spatially varying agronomic processes.

$$y(\mathbf{s}_i) = \beta_0(\mathbf{s}_i) + \sum_{k=1}^{p} \beta_k(\mathbf{s}_i) x_k(\mathbf{s}_i) + \varepsilon(\mathbf{s}_i) \tag{80}$$

In equation (80), $y(\mathbf{s}_i)$ represents the target variable at location $\mathbf{s}_i$, $x_k(\mathbf{s}_i)$ are the predictor variables, $\beta_k(\mathbf{s}_i)$ are spatially varying regression coefficients, and $\varepsilon(\mathbf{s}_i)$ is a random error term [39].

Haghighattalab et al. [55] used GWRs to predict grain yield on a plot level by examining the relationships between information derived from unmanned aerial systems (UAS) imagery and the grain yield. They found high correlation between measured phenotypic traits derived from imagery and grain yield ($\rho = 0.74$ and $\rho = 0.46$ [$p < 0.001$] for drought and irrigated environments, respectively). Residuals from GWR models were lower and less spatially dependent than methods using principal component regression, suggesting the superiority of spatially corrected models. Evans et al. [39] investigated the use of GWR for analysis of data from large on-farm experiments.

9.6 Multiscale Spatial Tree Models

Multiscale Spatial Tree Models decompose the spatial domain hierarchically into nested regions and fit predictive models at each level of the hierarchy [65]. These models support scale transitions by aggregating or disaggregating local predictions across tree levels [67]. Their structure makes them suitable for applications where both global trends and local variations must be represented simultaneously [41, 75, 123].

$$y(\mathbf{s}) = \sum_{m \in \mathcal{T}} \phi_m(\mathbf{s}) \theta_m + \varepsilon(\mathbf{s}) \tag{81}$$

Here, $y(\mathbf{s})$ is the observed value at location $\mathbf{s}$, $\phi_m(\mathbf{s})$ are spatial basis functions defined over tree node $m \in \mathcal{T}$, θ_m are scale-specific coefficients, and $\varepsilon(\mathbf{s})$ is the model error.

Zhu et al. [123] applied the multi-scale method by combining soil coring, penetrometer, and topographic data to map the depth-to-till on a 10-m resolution with soil coring and soil electrical conductivity measurements to map field capacity on a 20-m resolution. They evaluated different combinations of datasets, finding that in one data example where three data layers were present, the multi-scale model performed slightly better than regression kriging and cokriging; whereas in the other data example where two data layers were present, the multi-scale model produced results similar to regression kriging, but both outperformed cokriging, probably because soil core measurements were too sparse for cokriging to perform well. In situations where more than two data sources are available for mapping purposes, the multi-scale model showed better results.

9.7 Bayesian data fusion and hierarchical modeling

Bayesian hierarchical models provide a flexible probabilistic framework for spatial data fusion [8, 30].

A hierarchical formulation separates:

- **The latent spatial process:** the unobserved environmental field that one ultimately seeks to infer.
- **The observation process:** the mechanism through which sensors or measurements observe the latent field.
- **Uncertainty propagation:** probabilistic treatment of noise, measurement error, and incomplete sampling.

A generic formulation is:

$$y_i = A_i f + \varepsilon_i, \tag{82}$$

where:

- f is the latent spatial field;
- A_i is an observation operator describing how sensor i measures the field;
- ε_i denotes sensor-specific noise or measurement uncertainty.

This framework naturally accommodates:

- **Heterogeneous sensors:** different observation systems may possess distinct resolutions and error structures.
- **Multiscale support:** measurements collected at different supports can be represented consistently through observation operators.
- **Missing observations:** incomplete sampling can be incorporated probabilistically rather than treated as a purely technical inconvenience.
- **Uncertainty propagation:** posterior distributions provide uncertainty estimates directly linked to both observations and model assumptions [8, 49].

Bayesian fusion methods are increasingly important in climate science, remote sensing, and environmental forecasting because they provide a coherent framework for integrating physical knowledge and uncertain observations simultaneously [8, 91]. Bayesian Hierarchical Models (BHMs) provide a coherent probabilistic framework for modeling spatial data collected at multiple resolutions [74]. These models explicitly account for uncertainties and allow for the integration of diverse data sources by separating the data level, the process level, and the parameter level [110].

$$y_i \sim \mathcal{N}(z(\mathbf{s}_i), \sigma_i^2) \tag{83}$$

$$z(\mathbf{s}) = \mu(\mathbf{s}) + w(\mathbf{s}) \tag{84}$$

$$\mu(\mathbf{s}) = \mathbf{x}(\mathbf{s})^\top \boldsymbol{\beta}, \quad w(\mathbf{s}) \sim \mathrm{GP}(0, C(\mathbf{s}, \mathbf{s}')) \tag{85}$$

In equations (83)–(85), y_i are the observations, $z(\mathbf{s})$ is the latent spatial process, $\mu(\mathbf{s})$ is a mean function determined by covariates $\mathbf{x}(\mathbf{s})$ and coefficients $\boldsymbol{\beta}$, $w(\mathbf{s})$ is a spatially correlated random effect, and σ_i^2 is the observational error variance.

Stojanović et al. [110] evaluated the association between leaf area index and the spectral reflectance at various wavelengths by using spline-based kernel functions embedded into Bayesian Hierarchical Models. They compares models with three different levels of hierarchy: a non-hierarchical baseline model, a model with hierarchical bias parameter, and a model in which bias and kernel parameters are hierarchically structured. The authors considered using Bayesian Hierarchical Models yields qualitative benefits regarding interpretability, model complexity, and robustness.

9.8 Machine learning for spatial fusion

Machine learning methods have substantially expanded the capabilities of spatial data fusion [21, 32].

Deep learning architectures can integrate:

- **Raster imagery:** convolutional structures can extract spatial patterns from satellite or aerial data.
- **Spatial-temporal sequences:** recurrent and transformer architectures can model evolving dynamics across space and time.
- **Graph structures:** graph neural networks support irregular spatial relationships and network-based systems.
- **Physical simulations:** numerical model outputs can be incorporated jointly with observations.
- **Heterogeneous sensor streams:** multimodal architectures can combine distinct forms of spatial information simultaneously.

Attention mechanisms and transformer architectures additionally permit adaptive weighting of multiple information sources according to their contextual relevance [117].

These methods are powerful because they can learn highly nonlinear relationships without requiring explicit covariance assumptions [51, 80].

However, purely data-driven fusion methods may still struggle with [47, 98]:

- **Uncertainty quantification:** many deep models provide point predictions without rigorous probabilistic interpretation.
- **Physical interpretability:** learned representations may be difficult to connect with known environmental mechanisms.
- **Extrapolation outside training conditions:** predictive performance may degrade substantially in unseen spatial or climatic regimes.
- **Sparse observational regimes:** deep learning models may require more data than are available in many environmental applications.

These limitations motivate hybrid frameworks combining [20, 71]:

- physical models;
- geostatistics;
- probabilistic inference;
- machine learning.

9.9 Data fusion as operator integration

From a functional analytic perspective, data fusion can be interpreted as the integration of multiple observation operators acting on a common latent spatial field [38, 70].

Let

$$A_1, \ldots, A_m \tag{86}$$

represent distinct sensing operators [56, 111].

The fused inverse problem becomes:

$$y_i = A_i f + \varepsilon_i, \quad i = 1, \ldots, m. \tag{87}$$

This formulation is conceptually important because different datasets do not observe the same object in the same way [30, 54].

Each operator may:

- sample different spatial scales;
- introduce different smoothing effects;
- possess different uncertainty structures;
- observe different physical manifestations of the latent field.

This viewpoint unifies:

- **Multivariate geostatistics:** covariance-based integration of multiple spatial variables;
- **Inverse problems:** reconstruction of latent fields from indirect observations;
- **Bayesian hierarchical models:** probabilistic integration of heterogeneous information sources;
- **Machine learning fusion architectures:** nonlinear approximation of complex multimodal relationships.

Consequently, spatial data fusion is not merely a practical procedure for combining datasets. It is fundamentally an operator-theoretic problem defined over multiscale functional spaces [38, 64].

This interpretation becomes increasingly important as environmental datasets grow in dimensionality, heterogeneity, and spatial-temporal complexity [21, 32].

10 Hybrid geostatistics and machine learning

10.1 Beyond the false dichotomy

Geostatistics and machine learning are often presented as competing approaches to spatial prediction. This interpretation is misleading because the two frameworks address different components of the same inference problem [30, 60, 71].

Both approaches seek to infer unknown spatial functions from incomplete observations, but they emphasize different mathematical structures and different sources of information [52, 58].

Geostatistics focuses primarily on:

- **Stochastic spatial dependence:** nearby observations are treated as statistically related through covariance or variogram structures. The spatial configuration of the data is therefore part of the model itself.
- **Covariance structure:** spatial continuity is modeled explicitly, allowing the analyst to characterize spatial scales, anisotropy, and local uncertainty.
- **Uncertainty quantification:** prediction is accompanied by probabilistic information such as kriging variance or conditional realizations, which are essential for risk-sensitive applications.
- **Probabilistic interpolation:** predictions are derived from stochastic assumptions concerning the spatial field rather than only from empirical predictive accuracy.

Machine learning emphasizes:

- **Nonlinear approximation:** flexible algorithms can capture highly nonlinear relationships between predictors and responses without requiring explicit parametric assumptions.
- **High-dimensional predictors:** modern ML methods can integrate large numbers of environmental variables, raster layers, temporal features, and remote sensing products simultaneously.
- **Representation learning:** deep architectures can automatically extract latent structures and hierarchical features from complex spatial inputs.
- **Predictive scalability:** many ML methods remain computationally practical for datasets that are too large for classical covariance-based methods.

From a functional analytic perspective, these approaches are not mutually exclusive. They correspond to complementary approximation strategies acting on spatially structured function spaces [96, 105].

This observation is important because many environmental systems contain both deterministic nonlinear structure and latent stochastic spatial dependence. Ignoring either component may produce incomplete inference [43, 61].

This complementarity motivates hybrid spatial models [43, 61, 62, 118].

10.2 Residual spatial structure and decomposition

A central idea in hybrid modeling is that deterministic predictors rarely explain all spatial variability [60, 63].

Let

$$Z(\mathbf{s}) = f(\mathbf{x}(\mathbf{s})) + \varepsilon(\mathbf{s}), \tag{88}$$

where:

- $f(\mathbf{x}(\mathbf{s}))$ captures deterministic nonlinear relationships between predictors and the target variable;
- $\varepsilon(\mathbf{s})$ represents residual spatial dependence that remains unexplained after accounting for the predictors.

This decomposition is conceptually important because environmental systems are usually driven by multiple mechanisms simultaneously [52, 120].

Part of the variability may be explained by observable covariates such as:

- elevation;
- land cover;
- soil properties;
- climate variables;
- remote sensing products.

Another component may arise from latent spatial processes that are not directly observed, including:

- unresolved environmental heterogeneity;
- missing predictors;
- local transport mechanisms;
- measurement aggregation effects;
- multiscale stochastic variability.

Machine learning models may approximate the deterministic component effectively while still leaving residual spatial autocorrelation [88, 98].

Geostatistics can then model the remaining stochastic structure through covariance analysis [30, 52].

This decomposition is mathematically important because it separates:

- **Nonlinear covariate effects:** deterministic variation explained by observable environmental controls;

- **Latent spatial dependence:** structured residual variability associated with unresolved spatial organization;
- **Uncertainty propagation:** probabilistic characterization of the unexplained component of the field.

This interpretation also clarifies why adding coordinates directly as predictors is not equivalent to modeling covariance explicitly. Coordinates may help prediction numerically while still failing to provide a coherent stochastic representation of spatial dependence [30, 98].

10.3 Regression kriging and residual kriging

One of the most widely used hybrid methods is regression kriging [60, 62].

The prediction becomes:

$$\hat{Z}(\mathbf{s}) = \hat{f}_{\mathrm{ML}}(\mathbf{x}(\mathbf{s})) + \hat{\varepsilon}_{\mathrm{OK}}(\mathbf{s}), \tag{89}$$

where:

- $\hat{f}_{\mathrm{ML}}$ is a machine learning predictor responsible for capturing deterministic structure;
- $\hat{\varepsilon}_{\mathrm{OK}}$ is the kriged residual field, responsible for modeling remaining spatial autocorrelation.

This formulation combines:

- **Nonlinear predictive flexibility:** machine learning captures complex interactions that may be difficult to express through classical parametric models;
- **Explicit spatial covariance modeling:** kriging preserves information about spatial continuity and local dependence;
- **Probabilistic uncertainty representation:** residual kriging provides interpretable uncertainty estimates linked to spatial configuration and covariance structure.

Residual kriging is especially effective when:

- **Environmental covariates explain large-scale variation:** broad deterministic trends can be learned through regression or machine learning components;
- **Residual spatial autocorrelation persists locally:** remaining dependence can then be modeled separately through geostatistical methods;
- **Uncertainty quantification remains operationally important:** risk-sensitive applications require more than point prediction alone.

In practice, regression kriging often improves both predictive accuracy and spatial realism because the deterministic and stochastic components are modeled separately rather than forced into a single framework.

10.4 Gaussian processes and kernel unification

Gaussian process regression provides one of the clearest theoretical bridges between geostatistics and machine learning [96, 108].

A Gaussian process defines a probability distribution over functions:

$$f(\mathbf{x}) \sim \mathcal{GP}\left(m(\mathbf{x}), K(\mathbf{x}, \mathbf{x}')\right). \tag{90}$$

The covariance kernel

$$K(\mathbf{x}, \mathbf{x}') \tag{91}$$

plays the same mathematical role as covariance functions in geostatistics [30, 96].

Under appropriate assumptions, Gaussian process regression and kriging become equivalent formulations differing mainly in:

- **Terminology:** geostatistics typically emphasizes variograms and stochastic fields, while machine learning emphasizes kernels and probabilistic function learning;
- **Prior interpretation:** Gaussian processes are usually framed within Bayesian inference, whereas kriging historically emerged from regionalized variable theory;
- **Computational implementation:** the underlying mathematics is closely related, but optimization and software ecosystems often differ substantially.

This equivalence is theoretically important because it demonstrates that covariance-based geostatistics and kernel-based machine learning share the same Hilbert-space foundations [6, 105, 121].

Kernel methods, kriging, and probabilistic machine learning therefore emerge as different expressions of a common approximation framework [96].

10.5 Spatial deep learning and operator learning

Recent developments extend hybridization beyond classical residual modeling [76, 81].

Neural operators and physics-informed neural networks learn mappings between functional spaces rather than finite-dimensional vectors [76, 81, 82].

This distinction is important because many environmental systems are naturally governed by partial differential equations:

$$\mathcal{L}u = f. \tag{92}$$

In these settings, the objective is not merely interpolation between points, but approximation of spatial-temporal dynamics constrained by physical laws [81, 93].

Examples include:

- **Atmospheric transport:** advection, diffusion, and turbulence generate highly structured spatial-temporal dependence;

- **Groundwater flow:** hydraulic processes depend on heterogeneous permeability fields and boundary conditions;
- **Ocean circulation:** large-scale currents interact with multiscale stochastic forcing and sparse observations;
- **Heat diffusion:** spatial propagation depends on physically constrained transport operators;
- **Contaminant propagation:** uncertainty evolves dynamically across space and time.

Hybrid operator-learning frameworks combine [20, 60]:

- **Physical constraints:** governing equations and conservation laws provide structural regularization;
- **Stochastic uncertainty:** probabilistic representations account for incomplete observations and unresolved variability;
- **Observational data:** measurements constrain the latent spatial field;
- **Machine learning representations:** neural architectures approximate highly complex nonlinear mappings.

This direction represents a transition from purely empirical interpolation toward physically informed spatial inference [76, 93].

10.6 Spatial validation and predictive realism

A major risk in hybrid modeling is validation bias caused by spatial autocorrelation [88, 98].

Random cross-validation frequently overestimates predictive skill because nearby observations are statistically dependent [30, 89].

A flexible model may therefore appear highly accurate simply because training and testing samples belong to the same local spatial regime.

Spatially blocked validation provides a more realistic assessment by separating training and testing regions geographically [98, 109, 115].

This approach is important because it evaluates whether the model can:

- **Generalize spatially:** prediction quality should remain stable in genuinely unobserved regions;
- **Capture transferable structure:** the model should learn environmental relationships rather than memorize local neighborhoods;
- **Support operational deployment:** environmental applications often require extrapolation across space and time.

This issue is especially important in hybrid models because highly flexible machine learning components may memorize residual spatial patterns without learning physically meaningful structure [47, 98].

Consequently, spatial validation should be interpreted as a necessary component of defensible spatial inference rather than merely a model selection procedure [98, 115].

10.7 Why hybrid models are increasingly important

Environmental systems are simultaneously:

- **Nonlinear:** responses often emerge from interacting physical, biological, and chemical processes;
- **Multiscale:** different mechanisms operate at local, regional, and continental scales simultaneously;
- **Spatially correlated:** nearby observations are statistically dependent through latent spatial continuity;
- **Partially observed:** sampling is always finite and often sparse relative to the complexity of the field;
- **Physically constrained:** conservation laws, transport equations, and boundary conditions restrict admissible behavior.

No single modeling framework fully captures all these properties [30, 51].

Hybrid methods are therefore increasingly attractive because they combine:

- **Nonlinear predictive flexibility:** machine learning captures complex environmental relationships;
- **Probabilistic uncertainty representation:** geostatistics provides interpretable uncertainty linked to spatial structure;
- **Spatial dependence modeling:** covariance analysis preserves information about continuity and scale;
- **Multiscale integration:** different components of variability can be represented simultaneously;
- **Physical interpretability:** hybrid frameworks can incorporate scientific constraints and mechanistic understanding.

From a mathematical standpoint, hybridization is not an ad hoc engineering solution. It is a natural consequence of combining different approximation operators acting on complex spatial fields [38, 64].

The future of spatial analysis will likely depend increasingly on these integrated frameworks, where stochastic dependence, nonlinear learning, physical modeling, and uncertainty propagation are treated as complementary components of the same inference problem [30, 71, 76].

11 Case Study: Synthetic Spatial Field with Random Forest, Ordinary Kriging, and Hybrid Regression Kriging

This case study is designed as a controlled spatial experiment. Its purpose is not to reproduce a particular field campaign, but to isolate the relative behavior of three modeling paradigms under a data-generating process where both deterministic covariate effects and residual spatial dependence are present. This setting is useful because it exposes the conditions under which a machine learning model, a geostatistical model, and a hybrid model succeed or fail.

11.1 Synthetic data generation

Let $D \subset \mathbb{R}^2$ denote a bounded spatial domain. We generate $n = 600$ spatial locations

$$\mathbf{s}_i = (x_i, y_i)^\top, \qquad i = 1, \dots, n, \tag{93}$$

sampled uniformly over the domain. The coordinate names x and y are used only to emphasize the two-dimensional geometry of the synthetic field; they play the same role as longitude and latitude in the real-data case study, but without implying a geographic coordinate system.

At each location we define a collection of covariates intended to mimic the structure of the original monthly dataset. The variable names were chosen to preserve the same semantic interpretation as the real application. For example, `elevation` represents a static terrain control, `t2m` and `rh2m` represent near-surface temperature and relative humidity, `ps` denotes surface pressure, `ws2m` and `wd2m` represent wind speed and direction, `allsky_sfc_sw_dwn` is a radiation-type predictor, and `satellite_chirps` and `satellite_merge` are two synthetic satellite rainfall proxies. These names are intentionally retained because they make the synthetic experiment directly comparable to the real-data workflow and keep the interpretation of the model outputs physically meaningful.

The target variable, denoted $Y(\mathbf{s})$, is constructed as

$$Y(\mathbf{s}) = m\Big(\mathbf{z}(\mathbf{s})\Big) + W(\mathbf{s}) + \varepsilon(\mathbf{s}), \tag{94}$$

where $m(\cdot)$ is a nonlinear deterministic component, $W(\mathbf{s})$ is a spatially correlated Gaussian residual field, and $\varepsilon(\mathbf{s})$ is independent measurement noise. In the synthetic design, the mean structure includes nonlinear dependence on elevation, temperature, humidity, wind speed, and the satellite proxies. The residual term $W(\mathbf{s})$ is generated from a covariance model of the form

$$\mathrm{Cov}\Big(W(\mathbf{s}_i), W(\mathbf{s}_j)\Big) = \sigma_W^2 \exp\left(-\frac{\|\mathbf{s}_i - \mathbf{s}_j\|}{a}\right) + \tau^2 \mathbb{I}(i = j), \tag{95}$$

where a controls the spatial range, σ_W^2 the sill, and τ^2 the nugget effect. This construction ensures that the target field contains a component that is learnable from covariates and another component that is best handled through geostatistical smoothing.

To make the case study more realistic, a few additional derived variables are included. For instance, `t2m_range` represents the local diurnal temperature range, `dewpoint_deficit` encodes atmospheric dryness, `satellite_gap` measures disagreement between two synthetic satellite products, and `rh_temp_interaction` introduces a nonlinear coupling between humidity and temperature. These engineered features give the Random Forest a nontrivial covariate space while still leaving a structured spatial residual for the kriging stage to recover.

11.2 Spatial partitioning and validation design

To avoid overly optimistic validation caused by spatial autocorrelation, the domain is partitioned into spatial blocks using k-means clustering on coordinates. The block labels are then used to perform a grouped train-test split, so that entire regions are held out rather than randomly sampled points. This is essential because random pointwise splitting would allow near-duplicate spatial information to appear in both training and testing sets, inflating the apparent predictive skill of all methods.

In the reported experiment, the dataset contains 600 points, with 458 used for training and 142 for testing. The held-out test set is therefore spatially separated from the training set, giving a more realistic assessment of extrapolation across the domain.

11.3 Modeling strategies

Three models are compared.

11.3.1 Random Forest

The first benchmark is a Random Forest regression model. It represents the machine learning perspective in which the response is learned from covariates without explicit spatial covariance modeling. Let $\mathbf{x}(\mathbf{s})$ denote the feature vector at location $\mathbf{s}$. The Random Forest estimator can be written abstractly as

$$\widehat{Y}_{RF}(\mathbf{s}) = f_{RF}\Big(\mathbf{x}(\mathbf{s})\Big), \tag{96}$$

where f_{RF} is an ensemble of decision trees. In the present case study, the predictor set includes satellite variables, meteorological variables, and engineered nonlinear features such as temperature range, dewpoint deficit, and satellite differences. This model captures nonlinear interactions effectively, but it does not explicitly represent spatial dependence.

11.3.2 Ordinary Kriging

The second benchmark is Ordinary Kriging, which uses only spatial coordinates and the response values. Under ordinary kriging, the random field is decomposed as

$$Y(\mathbf{s}) = \mu + \eta(\mathbf{s}), \tag{97}$$

with constant unknown mean μ and spatially correlated zero-mean residual $\eta(\mathbf{s})$. Prediction at an unsampled location $\mathbf{s}_0$ is obtained through the kriging estimator

$$\widehat{Y}_{OK}(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i Y(\mathbf{s}_i), \tag{98}$$

where the weights λ_i are chosen to minimize the prediction variance under unbiasedness constraints. In the implementation, kriging is applied to a log-transformed target to stabilize the marginal distribution:

$$Y^{\star}(\mathbf{s}) = \log\Big(1 + Y(\mathbf{s})\Big). \tag{99}$$

Ordinary Kriging is an appropriate reference model because it directly exploits spatial autocorrelation. However, it cannot represent nonlinear dependence on the covariates. In a field with strong deterministic structure, this limitation can be severe.

11.3.3 Hybrid regression kriging

The hybrid model combines the strengths of both paradigms. First, a Random Forest estimates the deterministic component:

$$\widehat{m}_{RF}(\mathbf{s}) = f_{RF}\Big(\mathbf{x}(\mathbf{s})\Big). \tag{100}$$

Then the residuals

$$r(\mathbf{s}) = Y(\mathbf{s}) - \widehat{m}_{RF}(\mathbf{s}) \tag{101}$$

are modeled geostatistically by Ordinary Kriging. The final predictor is

$$\widehat{Y}_{HYB}(\mathbf{s}) = \widehat{m}_{RF}(\mathbf{s}) + \widehat{r}_{OK}(\mathbf{s}). \tag{102}$$

A crucial detail is that the kriging stage is fit to out-of-fold residuals rather than in-sample residuals. This avoids leakage and prevents the residual field from being artificially underestimated. The hybrid model therefore becomes a genuine regression kriging method rather than a simple post hoc smoothing layer.

11.4 Predictive performance

Table 2 summarizes the predictive results on the held-out spatial test set.

Table 2: Predictive performance for the synthetic spatial case study.

Model	Test RMSE	Test MAE	Test R^2
Random Forest	7.9867	6.1210	0.7363
Ordinary Kriging	11.8792	9.3594	0.4166
RF + OK residuals	7.4535	5.8079	0.7703

The hybrid model achieves the best overall performance. Relative to the Random Forest baseline, the hybrid reduces RMSE by approximately 6.7% and improves R^2 from 0.7363 to 0.7703. Relative to Ordinary Kriging, the improvement is much larger, with RMSE reduced by approximately 37.3%. These gains are modest in absolute terms but important in a spatial inference setting because they indicate that the residual spatial component contains exploitable structure beyond what the covariates already explain.

The comparison also reveals an important methodological point. Ordinary Kriging performs substantially worse than the Random Forest in this synthetic design because the target field is not purely a function of spatial proximity; it is also driven by nonlinear covariate effects. Kriging can interpolate the spatial surface, but it cannot reconstruct the covariate-driven mean structure by itself. The Random Forest does better because it captures that deterministic component, but it still leaves residual spatial dependence. The hybrid model is strongest because it represents both sources of variation simultaneously.

11.5 Observed versus predicted behavior

Figures 1, 2, and 3 show the observed-versus-predicted relationships for Ordinary Kriging, Random Forest, and the hybrid model, respectively. The kriging scatterplot exhibits systematic shrinkage toward a narrow prediction band, which is expected when the model is driven primarily by spatial averaging and a stationary covariance structure. The Random Forest scatterplot is more dispersed around the 1:1 line and captures the mean response more effectively, but it still displays spatially structured error. The hybrid scatterplot is the closest to the identity line and shows the most balanced dispersion across the prediction range.

11.6 Residual structure and spatial diagnostics

The residual map for the hybrid model, shown in Figure 4, is useful for assessing whether substantial spatial structure remains after both the machine learning and geostatistical stages. In an ideal hybrid specification, the residuals should appear closer to spatial white noise than in either baseline model. The presence of localized positive and negative patches indicates that the hybrid still leaves some unresolved fine-scale variation, but the amplitude of this structure is reduced relative to a purely covariate-based model.

The permutation importance plot shown in Figure 5 indicates that the most influential predictors are the satellite proxy variables and elevation. This is consistent with the data-generating process, which explicitly embeds strong dependence on those terms. The importance ranking is therefore not merely descriptive; it confirms that the synthetic experiment is behaving as intended and that the Random Forest is recovering the dominant deterministic structure.

11.7 Interpretation

This case study illustrates the central claim of the chapter in a concrete setting. The synthetic field contains both a nonlinear mean structure and a spatially correlated residual process. A single modeling family is not sufficient to capture both components optimally.

Random Forest is strong at nonlinear regression from heterogeneous covariates. Ordinary Kriging is strong at exploiting spatial continuity. The hybrid model is superior because it

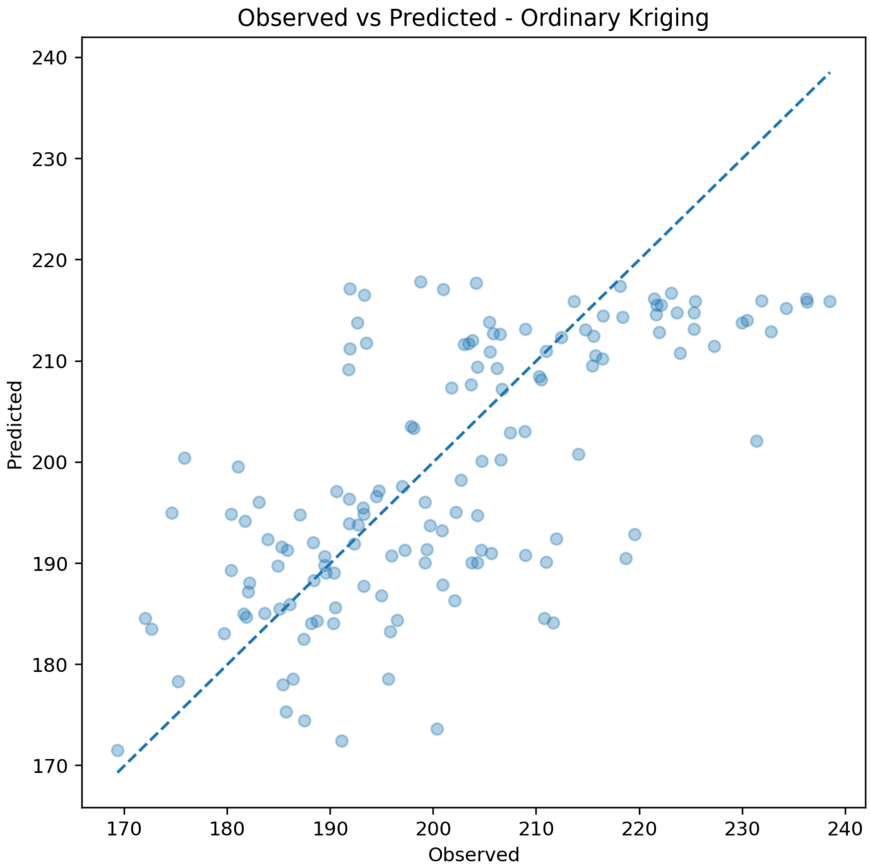

Figure 1: Observed versus predicted values for Ordinary Kriging on the synthetic test set.

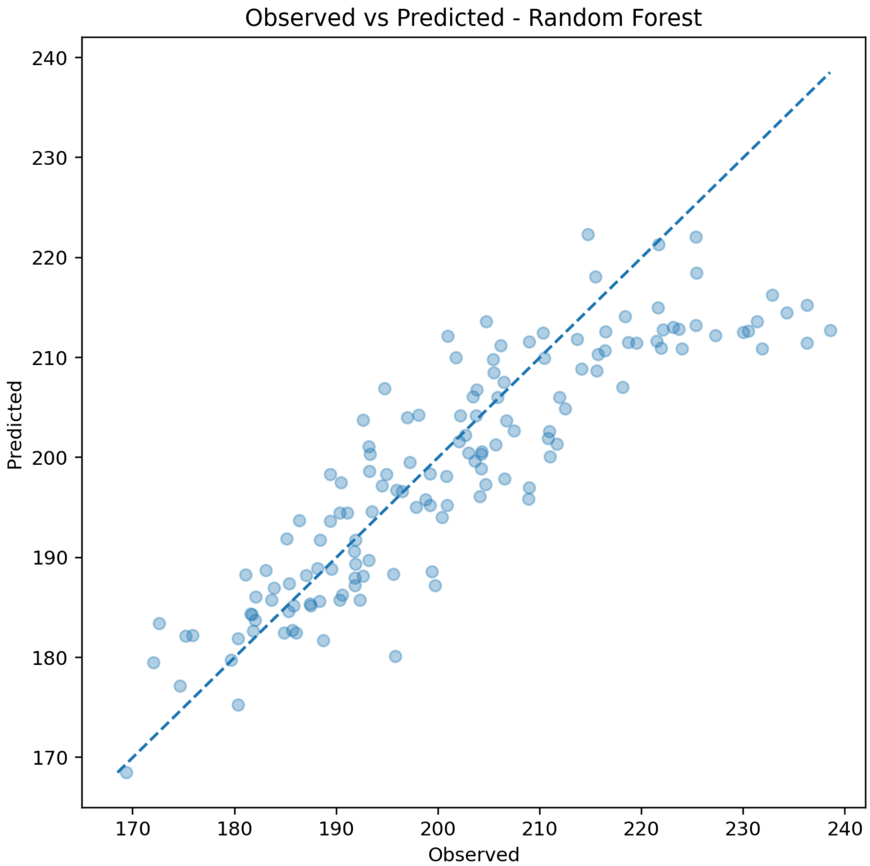

Figure 2: Observed versus predicted values for Random Forest on the synthetic test set.

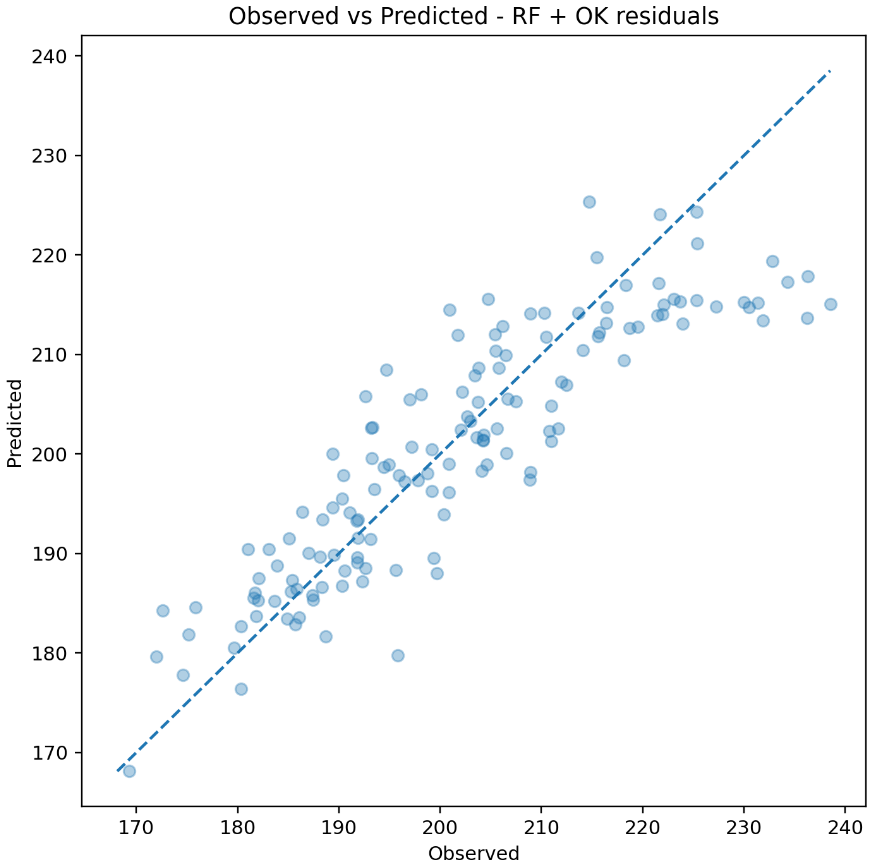

Figure 3: Observed versus predicted values for the hybrid RF + OK residual model on the synthetic test set.

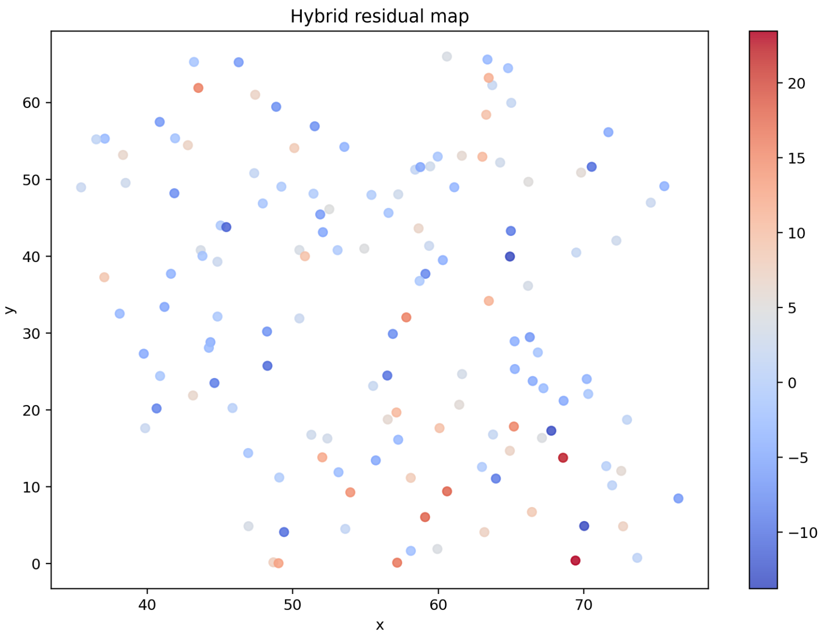

Figure 4: Spatial distribution of residuals for the hybrid model on the synthetic test set.

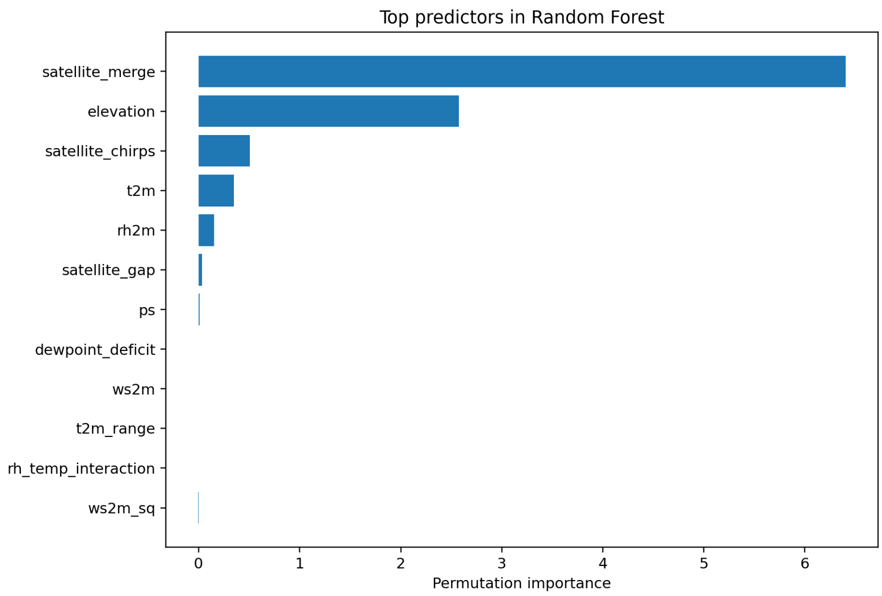

Figure 5: Permutation importance for the Random Forest baseline on the synthetic training/test split.

decomposes the problem into a deterministic part and a residual spatial part, then models each with the method best suited to it. This is the most important conceptual outcome of the experiment.

The example also clarifies why a hybrid approach can outperform either baseline even when the gain is modest. In spatial analysis, small improvements in RMSE and R^2 often correspond to a meaningful reduction in structural error, especially when the model is intended for inference, uncertainty propagation, or decision support rather than only point prediction. The improvement from 7.9867 to 7.4535 in RMSE is therefore methodologically relevant, not just numerically visible.

Finally, the synthetic design provides a controlled benchmark for the broader discussion in this chapter. Because the generating mechanism is known, the example makes it possible to see exactly what each method is learning: Random Forest learns the covariate response, Ordinary Kriging learns the spatial covariance, and the hybrid learns both. This is precisely the type of decomposition that motivates spatial data fusion, residual kriging, and hybrid geostatistical machine learning in real applications.

12 Concluding remarks

Spatial modeling should not be reduced to interpolation. At its core, it is the problem of inferring spatially structured functions from incomplete, noisy, and multiscale observations. Once the problem is framed this way, the apparent divide between geostatistics and machine learning becomes less important than the mathematical structure they share.

Geostatistics provides a rigorous probabilistic language for:

- **Spatial dependence:** covariance and variogram models describe how similarity changes with distance and direction.

- **Uncertainty propagation:** kriging variance, conditional simulation, and probabilistic thresholds provide interpretable measures of confidence.
- **Multiscale stochastic variability:** nested structures and coregionalization allow the decomposition of spatial variation across different scales.
- **Support-aware inference:** geostatistical methods can explicitly account for the fact that point values, pixels, and block averages are not interchangeable.

Machine learning provides a complementary language for:

- **Nonlinear approximation:** flexible models can capture complex relationships that are difficult to express with fixed covariance forms.
- **High-dimensional covariates:** remote sensing, topography, climate, and ancillary data can be exploited jointly.
- **Representation learning:** deep architectures can discover nonlinear features from structured spatial inputs.
- **Large-scale prediction:** modern ML systems can handle very large datasets more efficiently than classical kriging in many settings.

The opposition between these two traditions is therefore mostly pedagogical. From a functional analytic perspective, both are approximation methods on structured function spaces. Kriging is a projection problem. Kernel methods are RKHS approximations. Regularized inverse problems solve constrained reconstruction tasks. Neural operators learn mappings between function spaces. These are different expressions of a shared mathematical theme.

This viewpoint has several consequences. First, it clarifies why geostatistics and machine learning can be combined effectively rather than used as competing dogmas. Second, it explains why uncertainty, scale, and support matter so much in spatial analysis. Third, it shows that the most useful model is often not the one that is conceptually pure, but the one that best matches the structure of the data and the decision problem.

In practical terms, the strengths and limitations of each framework are complementary:

- **Geostatistics is strongest when spatial correlation dominates:** it provides principled interpolation, interpretable covariance structure, and explicit uncertainty estimates.
- **Machine learning is strongest when nonlinear predictors dominate:** it handles complex interactions, heterogeneous inputs, and high-dimensional feature spaces.
- **Hybrid methods are strongest when both effects matter:** they combine the explanatory power of covariates with the residual structure captured by spatial dependence.

Hybrid models are increasingly important because environmental systems are simultaneously nonlinear, spatially correlated, partially observed, multiscale, and physically constrained. No single framework fully captures all of these properties. A purely geostatistical model may underuse rich covariates. A purely machine learning model may ignore residual dependence and produce weak uncertainty estimates. A hybrid model can often do both jobs more credibly.

The broader conclusion is that spatial analysis should be treated as functional inference over complex fields rather than as a choice between interpolation algorithms. This shift in perspective encourages models that are more faithful to the data-generating process and more useful for decision-making. It also suggests a direction for future work: methods that integrate stochastic dependence, nonlinear approximation, physical constraints, support effects, and uncertainty propagation within a single coherent framework.

In that sense, the field is moving from map construction toward mathematically informed spatial inference.

References

[1] Robert A Adams and John J F Fournier. *Sobolev Spaces.* Academic Press, 2 edition, 2003.

[2] Katerina Aggelopooulou, Annamaria Castrignanò, Theofanis Gemtos, and Daniela Benedetto. Delineation of management zones in an apple orchard in greece using a multivariate approach. *Computers and Electronics in Agriculture*, 90:119–130, 2013. doi: 10.1016/j.compag.2012.09.009.

[3] Evangelos Anastasiou, Annamaria Castrignanò, Konstantinos Arvanitis, and Spyros Fountas. A multi-source data fusion approach to assess spatial-temporal variability and delineate homogeneous zones: A use case in a table grape vineyard in Greece. *Science of the Total Environment*, 684:155–163, 2019. doi: 10.1016/j.scitotenv.2019.05.324.

[4] Anastasios N Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

[5] Luc Anselin. *Spatial Econometrics: Methods and Models.* Kluwer Academic Publishers, 1988.

[6] Nachman Aronszajn. Theory of reproducing kernels. *Transactions of the American Mathematical Society*, 1950.

[7] Richard C Aster, Brian Borchers, and Clifford H Thurber. *Parameter Estimation and Inverse Problems.* Elsevier, 3 edition, 2019.

[8] Sudipto Banerjee, Bradley P Carlin, and Alan E Gelfand. *Hierarchical Modeling and Analysis for Spatial Data.* CRC Press, 2 edition, 2014.

[9] Yoshua Bengio, Aaron Courville, and Pascal Vincent. Representation learning: A review and new perspectives. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(8):1798–1828, 2013.

[10] Rishi Bommasani, Drew A Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.

[11] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[12] Leo Breiman, Jerome H Friedman, Richard A Olshen, and Charles J Stone. *Classification and Regression Trees*. Wadsworth, 1984.

[13] Haim Brezis. *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. Springer, 2011.

[14] Christopher F Brown, Steven P Brumby, Bin Chen, Shailesh Chaudhari, Jiquan Chen, Bryan Culbertson, John Dwyer, Saurabh Jha, Thomas F Keenan, Matthew Kelly, et al. Alphaearth foundations: An embedding field model for accurate and efficient global mapping from sparse label data. *arXiv preprint arXiv:2507.22291*, 2025.

[15] Tom B Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901, 2020.

[16] Chris Brunsdon, Stewart Fotheringham, and Martin Charlton. Geographically weighted regression. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 47(3):431–443, 1998.

[17] G Buttafuoco, R Quarto, F Quarto, M Conforti, A Venezia, C Vitti, and A Castrignanò. Taking into account change of support when merging heterogeneous spatial data for field partition. *Precision Agriculture*, 22:586–607, 2021. doi: 10.1007/s11119-020-09781-9.

[18] Guofeng Cao, Eun Hye Yoo, and Shaowen Wang. A statistical framework of data fusion for spatial prediction of categorical variables. *Stochastic Environmental Research and Risk Assessment*, 28:1785–1799, 2014. doi: 10.1007/s00477-013-0842-7.

[19] Annamaria Castrignanò and Gabriele Buttafuoco. Data processing. In *Agricultural Internet of Things and decision support for precision smart farming*, pages 139–182. Academic Press, Nova Iorque, 2020. doi: 10.1016/B978-0-12-818373-1.00003-2.

[20] Annamaria Castrignanò, Antonella Belmonte, Ilaria Antelmi, Ruggiero Quarto, Francesco Quarto, Sameh Shaddad, Valentina Sion, Maria Rita Muolo, Nicola A. Ranieri, Giovanni Gadaleta, Edoardo Bartoccetti, Carmela Riefolo, Sergio Ruggieri, and Franco Nigro. A geostatistical fusion approach using UAV data for probabilistic estimation of Xylella fastidiosa subsp. pauca infection in olive trees. *Science of the Total Environment*, 752:141814, 2021. doi: 10.1016/j.scitotenv.2020.141814.

[21] Ni Bin Chang and Kaixu Bai. *Multisensor data fusion and machine learning for environmental remote sensing*. CRC Press, Boca Raton, 2018. doi: 10.1201/b20703.

[22] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[23] Jean-Paul Chiles and Pierre Delfiner. *Geostatistics: modeling spatial uncertainty*, volume 713. John Wiley & Sons, Nova Iorque, 2012.

[24] Clay Team. Clay foundation model. *arXiv preprint arXiv:2410.18647*, 2024.

[25] Andrew D Cliff and J Keith Ord. *Spatial Processes: Models and Applications*. Pion, 1981.

[26] Alexis Comber, Christopher Brunsdon, Martin Charlton, Guanpeng Dong, Richard Harris, Binbin Lu, Yihe Lü, Daisuke Murakami, Tomoki Nakaya, Yunqiang Wang, et al. A route map for successful applications of geographically weighted regression. *Geographical Analysis*, 55(1):155–178, 2023.

[27] Yezhen Cong, Samir Khanna, Chenlin Meng, Patrick Liu, Ehsan Rozi, Yue He, Marshall Burke, David Lobell, and Stefano Ermon. Satmae: Pre-training transformers for temporal and multi-spectral satellite imagery. In *Advances in Neural Information Processing Systems*, volume 35, pages 197–211, 2022.

[28] John B Conway. *Functions of one complex variable II*. Springer Science & Business Media, 2012.

[29] Noel Cressie. Fitting variogram models by weighted least squares. *Journal of the international Association for mathematical Geology*, 17(5):563–586, 1985. doi: 10.1007/BF01032109.

[30] Noel Cressie. *Statistics for Spatial Data*. Wiley, 1993.

[31] Noel A Cressie. Change of support and the modifiable areal unit problem. *Geographical Systems*, 3(2-3):159–180, 1996.

[32] M. Dalla Mura, S. Prasad, F. Pacifici, P. Gamba, J. Chanussot, and J. A. Benediktsson. Challenges and Opportunities of Multimodality and Data Fusion in Remote Sensing. In *Proceedings of the IEEE*, volume 103, Piscataway, 2015. Institute of Electrical and Electronics Engineers. doi: 10.1109/JPROC.2015.2462751.

[33] Ingrid Daubechies. *Ten Lectures on Wavelets*. SIAM, 1992.

[34] Clayton V Deutsch and Andre G Journel. *Geostatistical software library and user's guide*, volume 119. Elsevier, Nova Iorque, 1992.

[35] Clayton V Deutsch and André G Journel. *GSLIB: Geostatistical Software Library and User's Guide*. Oxford University Press, 2 edition, 1998.

[36] Peter J Diggle and Paulo J Ribeiro. *Model-Based Geostatistics*. Springer, 2007.

[37] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

[38] Heinz Engl, Martin Hanke, and Andreas Neubauer. *Regularization of Inverse Problems*. Springer, 1996.

[39] Fiona H Evans, Angela Recalde Salas, Suman Rakshit, Craig A Scanlan, and Simon E Cook. Assessment of the use of geographically weighted regression for analysis of large on-farm experiments and implications for practical application. *Agronomy*, 10(11):1720, 2020.

[40] D. Fasbender and P. Bogaert. Data fusion in a spatial multivariate framework: Trading off hypotheses against information. In Amílcar Soares, Maria João Pereira, and Roussos Dimitrakopoulos, editors, *geoENV VI – Geostatistics for Environmental Applications*, pages 457—-466. Springer, Nova Iorque, 2008.

[41] Marco AR Ferreira and Herbert KH Lee. *Multiscale modeling: a Bayesian perspective*. Springer, 2007.

[42] A Stewart Fotheringham, Chris Brunsdon, and Martin Charlton. Geographically weighted regression. *The Sage handbook of spatial analysis*, 1:243–254, 2009.

[43] Francky Fouedjio and Jens Klump. Exploring prediction uncertainty of spatial data in geostatistical and machine learning approaches. *Environmental Earth Sciences*, 78(1):38, 2019. doi: 10.1007/s12665-018-8032-z.

[44] Jerome Friedman and Bogdan E Popescu. Gradient directed regularization. *Unpublished manuscript, http://www-stat. stanford. edu/jhf/ftp/pathlite. pdf*, 2004.

[45] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning*, pages 1050–1059, 2016.

[46] Bo-Cai Gao. Ndwi—a normalized difference water index for remote sensing of vegetation liquid water from space. *Remote Sensing of Environment*, 58(3):257–266, 1996.

[47] Jakob Gawlikowski, Cedrique Rovile Njieutcheu Tassi, Mohsin Ali, Jongseok Lee, Matthias Humt, Jianxiang Feng, Anna Kruspe, Rudolph Triebel, Peter Jung, Ribana Roscher, et al. A survey of uncertainty in deep neural networks. *Artificial Intelligence Review*, 56:1513–1589, 2023.

[48] Peter V Gehler, Bernhard Schölkopf, G Camps-Valls, and L Bruzzone. An introduction to kernel learning algorithms. *Kernel methods for remote sensing data analysis*, pages 25–48, 2009.

[49] Alan E Gelfand, Li Zhu, and Bradley P Carlin. On the change of support problem for spatio-temporal data. *Biostatistics*, 2(1):31–45, 2001.

[50] Tilmann Gneiting. Nonseparable, stationary covariance functions for space-time data. *Journal of the American Statistical Association*, 97(458):590–600, 2002.

[51] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.

[52] P. Goovaerts. *Geostatistics for natural reources evaluation.* Oxford University Press, Nova Iorque, 1997.

[53] Pierre Goovaerts. Factorial kriging analysis: a useful tool for exploring the structure of multivariate spatial soil information. *Journal of soil science*, 43(4):597–619, 1992. doi: 10.1111/j.1365-2389.1992.tb00163.x.

[54] C. A. Gotway and L. J. Young. Combining incompatible spatial data. *Journal of the American Statistical Association*, 97(458):632–648, 2002. doi: 10.1198/016214502760047140.

[55] Atena Haghighattalab, Jared Crain, Suchismita Mondal, Jessica Rutkoski, Ravi Prakash Singh, and Jesse Poland. Application of geographically weighted regression to improve grain yield prediction from unmanned aerial system imagery. *Crop Science*, 57(5):2478–2489, 2017.

[56] D L Hall and S A H McMullen. *Mathematical Techniques in Multisensor Data Fusion.* Artech House Publishers, Londres, 2 edition, 2004.

[57] Robert M Haralick, K Shanmugam, and Its'hak Dinstein. Textural features for image classification. *IEEE Transactions on Systems, Man, and Cybernetics*, SMC-3 (6):610–621, 1973.

[58] Trevor Hastie, Robert Tibshirani, Jerome H Friedman, and Jerome H Friedman. *The elements of statistical learning: data mining, inference, and prediction*, volume 2. Springer, Nova Iorque, 2009.

[59] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16000–16009, 2022.

[60] T Hengl, GBM Heuvelink, and DG Rossiter. About regression-kriging: from theory to interpretation of results. *Computers & Geosciences*, 33(10):1301–1315, 2007.

[61] Tomislav Hengl, Branislav Bajat, Dragan Blagojević, and Hannes I Reuter. Geostatistical modeling of topography using auxiliary maps. *Computers & Geosciences*, 34 (12):1886–1899, 2008.

[62] Tomislav Hengl, Henk Sierdsema, Andreja Radović, and Arta Dilo. Spatial prediction of species' distributions from occurrence-only records: combining point pattern analysis, enfa and regression-kriging. *Ecological modelling*, 220(24):3499–3511, 2009.

[63] Tomislav Hengl, Madlene Nussbaum, Marvin N Wright, Gerard BM Heuvelink, and Benedikt Gräler. Random forest as a generic framework for predictive modeling of spatial and spatio-temporal variables. *PeerJ*, 6:e5518, 2018. doi: 10.7717/peerj.5518.

[64] Tailen Hsing and Randall Eubank. *Theoretical Foundations of Functional Data Analysis, with an Introduction to Linear Operators.* Wiley, 2015.

[65] Hsin-Cheng Huang and Noel Cressie. Multiscale spatial modeling. In *1997 Proceedings of the Section on Statistics and the Environment*, pages 49–54, 1997.

[66] Alfredo R Huete. A soil-adjusted vegetation index (savi). *Remote Sensing of Environment*, 25(3):295–309, 1988.

[67] William W Irving, Paul W Fieguth, and Alan S Willsky. An overlapping tree approach to multiscale stochastic modeling and estimation. *IEEE Transactions on Image Processing*, 6(11):1517–1529, 1997.

[68] Johannes Jakubik, Sujit Roy, Christopher E Phillips, Paolo Fraccaro, Denys Godwin, Bianca Zadrozny, Daniela Szwarcman, Carlos Gomes, Gabby Nyirjesy, Blair Edwards, et al. Foundation models for generalist geospatial artificial intelligence. *arXiv preprint arXiv:2310.18660*, 2023.

[69] André G Journel and Mario E Rossi. When do we need a trend model in kriging? *Mathematical Geology*, 21(7):715–739, 1989.

[70] Jari Kaipio and Erkki Somersalo. *Statistical and Computational Inverse Problems*. Springer, 2006.

[71] Mikhail Kanevski. *Machine learning for spatial environmental data: theory, applications, and software*. EPFL press, Londres, 2009.

[72] Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 2017.

[73] Roger Koenker and Kevin F Hallock. Quantile regression. *Journal of Economic Perspectives*, 15(4):143–156, 2001.

[74] Eric D Kolaczyk. Bayesian multiscale models for poisson processes. *Journal of the American Statistical Association*, 94(447):920–933, 1999.

[75] Eric D Kolaczyk and Haiying Huang. Multiscale statistical models for hierarchical spatial aggregation. *Geographical Analysis*, 33(2):95–118, 2001.

[76] Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to pdes. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

[77] Erwin Kreyszig. *Introductory Functional Analysis with Applications*. Wiley, 1989.

[78] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems*, volume 25, 2012.

[79] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[80] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. *Nature*, 521 (7553):436–444, 2015.

[81] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

[82] Yulong Lu, Jianfeng Lu, and Min Wang. A priori generalization analysis of the deep ritz method for solving high dimensional elliptic partial differential equations. In *Conference on learning theory*, pages 3196–3241. PMLR, 2021.

[83] Stéphane G Mallat. A theory for multiresolution signal decomposition: The wavelet representation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11(7):674–693, 1989.

[84] G. Matheron. Principles of geostatistics. *Economic Geology*, 58(8):1246—-1266, 1963. doi: 10.2113/gsecongeo.58.8.1246.

[85] Georges Matheron. *The Theory of Regionalized Variables and Its Applications*. Les Cahiers du Centre de Morphologie Mathématique, 1971.

[86] Stuart K McFeeters. The use of the normalized difference water index (ndwi) in the delineation of open water features. *International Journal of Remote Sensing*, 17(7): 1425–1432, 1996.

[87] Jeremy Mennis. Mapping the results of geographically weighted regression. *The Cartographic Journal*, 43(2):171–179, 2006.

[88] Hanna Meyer, Christoph Reudenbach, Stephan Wöllauer, and Thomas Nauss. Importance of spatial predictor variable selection in machine learning applications–moving from data reproduction to spatial prediction. *Ecological Modelling*, 411: 108815, 2019. doi: 10.1016/j.ecolmodel.2019.108815.

[89] P Moran. Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2):17—-23, 1950. doi: 10.2307/2332142.

[90] Guy P Nason and Bernard W Silverman. The stationary wavelet transform and some statistical applications. *Lecture Notes in Statistics*, 103:281–299, 1995.

[91] Hai Nguyen, Matthias Katzfuss, Noel Cressie, and Amy Braverman. Spatio-temporal data fusion for very large remote sensing datasets. *Technometrics*, 56(2):174–185, 2014.

[92] Jonne Pohjankukka, Tapio Pahikkala, Pasi Nevalainen, and Jukka Heikkonen. Estimating the prediction performance of spatial models via spatial k-fold cross validation. *International Journal of Geographical Information Science*, 31(10): 2001–2019, 2017.

[93] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[94] James O Ramsay and Bernard W Silverman. *Functional Data Analysis*. Springer, 2 edition, 2005.

[95] James O Ramsay and Bernard W Silverman. Principal components analysis for functional data. In *Functional data analysis*, pages 85–109. Springer, 2005.

[96] Carl Edward Rasmussen and Christopher Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[97] Jacques Rivoirard. *Introduction to disjunctive kriging and non-linear geostatistics*. Clarendon Press, Londres, 1994.

[98] David R Roberts, Volker Bahn, Simone Ciuti, Mark S Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.

[99] Galina L Rogova and Vincent Nimier. Reliability in information fusion: literature survey. In *Proceedings of the seventh international conference on information fusion*, volume 2, pages 1158–1165, Xangai, 2004. Institute of Electrical and Electronics Engineers.

[100] John W Rouse, Robert H Haas, John A Schell, and Donald W Deering. Monitoring vegetation systems in the great plains with erts. Technical report, NASA Goddard Space Flight Center, 1974.

[101] Leonid I Rudin, Stanley Osher, and Emad Fatemi. Nonlinear total variation based noise removal algorithms. *Physica D: Nonlinear Phenomena*, 60(1–4):259–268, 1992.

[102] Walter Rudin. *Functional Analysis*. McGraw-Hill, 2 edition, 1991.

[103] Paul D Sampson and Peter Guttorp. Nonparametric estimation of nonstationary spatial covariance structure. *Journal of the American Statistical Association*, 87 (417):108–119, 1992.

[104] Thomas J Santner, Brian J Williams, and William I Notz. *The Design and Analysis of Computer Experiments*. Springer, 2003.

[105] Bernhard Schölkopf and Alexander J Smola. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. MIT Press, 2002.

[106] César de Oliveira Ferreira Silva, Rodrigo Lilla Manzione, and Stanley Robson de Medeiros Oliveira. Exploring 20-year applications of geostatistics in precision agriculture in brazil: what's next? *Precision Agriculture*, 24:2293–2326, 2023. doi: 10.1007/s11119-023-10041-9.

[107] Cesar de Oliveira Ferreira Silva, Celia Regina Grego, Rodrigo Lilla Manzione, Stanley Robson de Medeiros Oliveira, Gustavo Costa Rodrigues, and Cristina Aparecida Gonçalves Rodrigues. Delineation of management zones dealing with low sampling and outliers. *Precision Agriculture*, 26(1):19, 2025.

[108] Michael L Stein. *Interpolation of Spatial Data: Some Theory for Kriging*. Springer, 1999.

[109] Andy Stock. Choosing blocks for spatial cross-validation: lessons from a marine remote sensing case study. *Frontiers in Remote Sensing*, 6:1531097, 2025.

[110] Olivera Stojanović, Bastian Siegmann, Thomas Jarmer, Gordon Pipa, and Johannes Leugering. Bayesian hierarchical models can infer interpretable predictions of leaf area index from heterogeneous datasets. *Frontiers in Environmental Science*, 9: 780814, 2022.

[111] Albert Tarantola. *Inverse Problem Theory and Methods for Model Parameter Estimation.* SIAM, 2005.

[112] Andrey N Tikhonov and Vasily Y Arsenin. *Solutions of Ill-Posed Problems.* Winston, 1977.

[113] Waldo R Tobler. A computer movie simulating urban growth in the detroit region. *Economic geography*, 46(sup1):234–240, 1970.

[114] Compton J Tucker. Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sensing of Environment*, 8(2):127–150, 1979.

[115] Roozbeh Valavi, Jane Elith, José J Lahoz-Monfort, and Gurutzeta Guillera-Arroita. blockcv: An r package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution*, 10(2):225–232, 2019.

[116] Vladimir N Vapnik. *Statistical Learning Theory.* Wiley, 1998.

[117] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[118] Kévin Vaysse and Philippe Lagacherie. Using quantile regression forest to estimate uncertainty of digital soil mapping products. *Geoderma*, 291:55–64, 2017. doi: 10.1016/j.geoderma.2016.12.017.

[119] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World.* Springer, 2005.

[120] H. Wackernagel. *Multivariate geostatistics: an introduction with applications.* Springer, Nova Iorque, 2003. doi: 10.2307/2291758.

[121] Grace Wahba. *Spline Models for Observational Data.* SIAM, 1990.

[122] Richard Webster and Margaret A Oliver. *Geostatistics for environmental scientists.* John Wiley & Sons, Nova Iorque, 2007.

[123] Jun Zhu, Cristine LS Morgan, John M Norman, Wei Yue, and Birl Lowery. Combined mapping of soil properties using a multi-scale tree-structured spatial model. *Geoderma*, 118(3-4):321–334, 2004.